



Research Paper

Acoustic sensing for investigating critical heat flux enhancement during pool boiling on electrodeposited copper foams

Hari Pandey^a, Hamed Mehrabi^b, Amanda Williams^a, Carolina Mira-Hernández^c, Robert H. Coridan^{b,d}, Han Hu^{a,b,*}

^a Department of Mechanical Engineering, University of Arkansas, Fayetteville, AR 72701, USA

^b Materials Science and Engineering Program, University of Arkansas, Fayetteville, AR 72701, USA

^c School of Engineering, Universidad Pontificia Bolivariana Medellín, Colombia

^d Department of Chemistry and Biochemistry, University of Arkansas, Fayetteville, AR 72701, USA

ARTICLE INFO

Keywords:

Pool boiling

Critical heat flux

Hierarchical copper foams

Acoustic emissions

Electrodeposition with pH control

ABSTRACT

Nano-engineered surfaces have shown great promise in improving boiling heat flux and heat transfer coefficient for thermal management. Electrodeposited copper foams have attracted lots of scientific and engineering interest owing to their multi-tier structures. In this study, hierarchical copper foams with ultra-wicking properties were fabricated using cathodic deposition at different pH, including both acidic (pH 0) and basic (pH 10 and 12) solutions. Pendant droplet wicking tests and pool boiling experiments were performed to characterize the wicking flux and the critical heat flux (CHF) of the fabricated copper foams, respectively. The CHF enhancement mechanism in hierarchical copper foams was investigated with high-speed optical imaging and acoustic emission (AE) sensing during both the wicking and boiling tests. The results show that the wicking flux of the electrodeposited copper foams increases with the pH due to smaller dendrites and crystallites. Boiling CHF is also higher on the base-deposited copper foams (pH 10 and pH 12) than on the acid-deposited foams (pH 0). Consistent acoustic signatures are observed between wicking and boiling, i.e., smaller AE amplitudes are observed at higher wicking flux and higher CHF, which supports wicking-based CHF enhancement theories. A spike in the AE amplitude is observed at CHF during water boiling on copper foam surfaces due to the rapid growth of the vapor layer blanketing the boiling surface. During the nucleate boiling regime, the AE amplitude decreases with the heat flux in the low heat flux stage but increases with the heat flux in the high heat flux stage. This relationship is found to be due to the interplay of bubble size and bubble count in the stress field that bubbles exert on the heater surface. By analyzing the maximum bubble size, bubble count, and the power and dominant frequency of vapor fraction at varying heat fluxes from high-speed videos, our work shows no obvious effect of the surface structures on the dynamics of the apparent liquid–vapor interface. Combining the optical and acoustic analysis, it is concluded that the key role that surface structures play in CHF enhancement is improving capillary wicking underneath the bubbles rather than affecting the dynamics of the apparent liquid–vapor interfaces. This study not only explores the relationship between electrochemical control and wicking properties but also demonstrates the role of wicking in structure-enhanced boiling CHF through acoustic emission analysis.

1. Introduction

Nucleate boiling is a highly efficient heat dissipation method and is central to electronics cooling [1,2], power generation [3], and distillation [4], among many other industrial applications. The critical heat flux (CHF) represents an upper limit of boiling heat transfer without causing a transition from nucleate boiling to film boiling, which is characterized by significantly higher thermal resistance due to the vapor film

blanketing the heater surface. Therefore, understanding and controlling CHF is crucial for ensuring safe and efficient heat transfer. The cause of CHF has been attributed to various physical mechanisms, including hydrodynamic instabilities [5,6], force balance [7], microlayer lift-off [8], microlayer dry-out [9], and bubble interactions [10,11]. Zuber's theory based on the Helmholtz instability of vapor columns leaving the heater surface has been shown to agree very well with experimental data using planar, wetting surfaces [5]. As such, the Zuber's limit has become

* Corresponding author.

E-mail address: hanhu@uark.edu (H. Hu).

<https://doi.org/10.1016/j.applthermaleng.2023.121807>

Received 13 June 2023; Received in revised form 8 October 2023; Accepted 16 October 2023

Available online 21 October 2023

1359-4311/© 2023 Elsevier Ltd. All rights reserved.

an important metric to benchmark boiling performance ($\sim 110 \text{ W/cm}^2$ for water at ambient pressure).

To meet the ever-increasing cooling demands in microelectronics and power systems, several strategies have been adopted to enhance boiling CHF beyond the Zuber's limit, e.g., engineering heater surfaces with micro or nanoscale textures and adding nano-additives to the working fluids [12–14]. Surface modification has been particularly promising due to the rapid advances in nanomanufacturing technologies in recent decades. Various textured surfaces have been designed, manufactured, and tested for boiling, including micropillar arrays [15–21], re-entrant cavities [1,22,23], other ordered [24–27] and disordered microporous coatings [28–31], nanowires [32–36] and hierarchical structures with textures across the scales [20,37–56]. Among these surfaces, metal foams have shown great potential to improve boiling CHF and heat transfer coefficient (HTC) due to their unique multi-tier textures [41,47–56]. For example, Li et al. demonstrated an ultra-high pool boiling CHF of 400 W/cm^2 and an HTC of $12 \text{ W/(cm}^2 \text{ K)}$ for water boiling on electrodeposited copper foams [41]. Parametric studies have been performed to reveal the role of porosity [56], pore density [47,49–51,54,56], foam thickness [41,48,50,51,54,56], thickness gradient [45,55], and wettability [52] on boiling CHF enhancements.

The fundamental understanding of CHF enhancement mechanisms is critical to enable the rational design of surface structures. To this end, many theoretical, empirical, and semi-empirical CHF models have been developed to explain and predict structure-enhanced CHF, e.g., amplified surface tension force induced by elongated contact line [19,39,57], increased microconvection due to rapid interface motion [58], delayed dry out due to capillary wicking [20,41,59,60], and structure-induced vapor confinement [61]. Among these algorithms, wicking-based theories have attracted a lot of scientific interest owing to the vastly available experimental evidence showing strong correlations between boiling CHF and wicking metrics. A variety of experimental approaches have been developed and used to characterize surface wickability, including radial wicking of sessile droplets [17,28,62–65], radial wicking of pendant droplets [20,41,66–68], and linear wicking of liquid films [69,70]. Synchrotron X-ray imaging has also been used to visualize the wicking front in surface structures [71,72]. Many different wicking metrics have been introduced to represent surface wickability, including phenomenological parameters such as early-stage wicking flux [20,41], volumetric wicking rate [66], wicking factor [17], and theoretical metrics such as the superhydrophilicity metric [67], propagation coefficient [69], rewetting time scale [35], etc. Besides correlating boiling performance with ex-situ measured wicking metrics, efforts have been made to understand bubble dynamics during structure-enhanced boiling through quantitative analysis of boiling images, particularly, high-speed videos that capture the high-frequency dynamics during boiling. By segmenting bubble and liquid–vapor interfaces from the images, characteristic properties of boiling can be extracted in quantifiable parametric data and interpreted with changes in CHF and HTC, including the bubble departure diameter [41,73–79], bubble departure frequency [41,78–81], nucleation site density [75,79,80,82], wetted area fraction [83–85], average bubble diameter [86,87], and void fraction [88].

There has been a growing interest in defining a universal wicking metric that can be directly used for CHF predictions. Rahman et al. experimentally demonstrated a linear relationship between boiling CHF and the wicking flux measured in pendant droplet radial wicking tests for a wide range of textured surfaces [20]. This linear relationship is also supported by data from other studies, e.g., water boiling on sintered copper surfaces covered by nanograss and high-density microcavities [28]. Nevertheless, Li et al.'s boiling data using electrodeposited copper foams suggested the relationship between CHF and wicking flux may be nonmonotonic and CHF is more closely related with bubble departure diameter and frequency at onsite of nucleate boiling [41]. Zhang et al. discussed improving boiling CHF by simultaneously strengthening liquid wicking and manipulating bubble behaviors using copper meshes with gradient thickness, yielding a CHF of $\sim 300 \text{ W/cm}^2$ and an HTC of

$34 \text{ W/(cm}^2 \text{ K)}$ [45]. Dhillon et al. used experimental data and theoretical modeling to demonstrate that CHF is related to the ratio of the rewetting and heating time scales [35]. Hu et al. developed a coupled wicking and evaporation CHF model that directly relates CHF with the dry spot size underneath the bubble, and the dry spot size is a result of the competition between capillary wicking and microlayer evaporation [60]. This model is further improved with a universal evaporation figure of merit defined for structured surfaces [89]. By using the scaling relationship between the thin film density and evaporation flux, Song et al. developed a unified descriptor for wicking-enhanced boiling CHF that combines the effect of both evaporation and capillary wicking in boiling [59]. Li et al. experimentally visualized the dry spot and the wicking region underneath bubbles during nucleate boiling using in-liquid endoscopy [90] and introduced a dimensionless liquid retention number (Ret) to characterize the contribution of liquid retention in the surface structures to CHF enhancement [91]. Despite the different assumptions and metrics adopted by these models, there is a consensus that boiling CHF cannot be solely determined by wicking, but instead, by the combined effects of evaporation and wicking. Since the optimal dimensions for evaporation are generally much smaller than that of capillary wicking (with the former representing the balance between vapor kinetics and nanoconfinement [92] and the latter representing the balance between capillary pressure and viscous resistance [60]), multi-tier structures such as metal foams with textures across the scales are most promising for higher CHF. This highlights that several studies recognize wicking in one way or the other, and a boiling surface must have strong wicking properties while understanding the role of wicking in boiling heat transfer or achieving a higher CHF.

In this paper, electrodeposition with pH control is used to fabricate multi-tier copper foam structures for higher surface wickability and enhanced thermal performance. The wicking and boiling performance of the copper foams fabricated in acidic solution (pH 0) and basic solution (pH 10 and 12) are experimentally characterized with a multimodal sensing system including temperature, acoustic, and optical signals. Three samples deposited at widely different pH (0, 10, and 12) were considered to generalize the overall effect of the pH-deposition environment over the wicking performance. The bubble size, bubble count, and vapor fraction are analyzed using high-speed optical images for copper foams fabricated under varying pH to examine the effect of surface wickability on bubble dynamics. Furthermore, we have recorded the acoustic emissions (AE) generated during both wicking and boiling vapor dynamics to probe the role of wicking in boiling CHF enhancement. Acoustic emissions are transient mechanical waves or vibrations that are generated by a sudden release of energy in materials under external force or internal state changes. AE sensing measures and analyzes dynamic behaviors through acoustic waves and has been used in boiling studies to identify regime transitions [93–96], probe bubble dynamics [97,98], and predict heat flux [99,100]. We extend the implementation of this non-intrusive sensing technology to wicking tests to identify acoustic features representing capillary wicking and compare them with AE signals captured during boiling. By introducing a parameter space for boiling surface fabrication (pH control) and a new angle of probing wicking in boiling (acoustic emission), our work not only advances the fundamental understanding of structure-enhanced boiling heat transfer but also endorses an exploration of a new direction: applying acoustics analysis to the structured surfaces beyond the state-of-the-art.

2. Methodology

2.1. Surface fabrication

Two different types of modified surfaces were used in the boiling experiment comprising a polished copper surface and hierarchical copper foams. The polished copper surface was prepared by polishing a 110 copper block (McMaster-Carr 89275K38) with fine sandpaper. Before

polishing, the boiling surface of the copper block was cleansed with chemicals in the following order: acetone, ethanol, isopropanol, and deionized water to remove any impurities. The treated surface was unidirectionally polished by flipping the heater block where the boiling surface touched the sandpaper (grit 320 followed by grit 600).

The hierarchical copper foams were fabricated using the electrodeposition method with pH control, following the recipes and operating procedures detailed in [101]. As illustrated in Fig. 1, copper surfaces were polished using grit 1000 sandpaper and then submerged into the prepared solutions for electrodeposition. The materials include water (HPLC grade; BDH), acetone (HPLC grade; VWR), methanol (HPLC grade; VWR), and isopropyl alcohol (HPLC grade; VWR), and were used as received. Copper (II) sulfate pentahydrate (99.99 %; Sigma Aldrich), sulfuric acid (96 %; Sigma Aldrich), sodium hydroxide (50 % (W/W) in water; Sigma Aldrich), lactic acid (95 %; Sigma Aldrich), which were also used as received. An Ag/AgCl reference electrode (stored in saturated KCl; BASi), was used during electrodeposition. High purity 0.25 mm-thick copper foil electrodes (99.9985 %; Alfa Aesar) and high purity copper blocks (110 copper, McMaster Carr) were used as electrodeposition substrates for wicking and boiling experiments, respectively. Three copper deposition solutions with pH values of 0, 10, and 12 were used to deposit copper dendrites. The pH 0 solution consisted of an aqueous solution of 0.40 M CuSO_4 and 1.5 M H_2SO_4 . Lactic acid was used to achieve stable copper ions in basic conditions [102]. The pH 10 and pH 12 solutions consisted of 0.4 M CuSO_4 and 3 M lactic acid, followed by the titration of a sufficient volume of concentrated NaOH to reach the desired pH as indicated by a pH meter. The dendritic copper foams were electrodeposited in a three-electrode electrochemical cell with an Ag/AgCl reference electrode, and a 3 cm^2 copper foil was used as the counter electrode. The pH 0 dendritic foams were deposited at a constant potential of -2.0 V vs. Ag/AgCl for 15 s. The pH 10 and pH 12 foams were deposited in their corresponding solutions at a constant potential of -2.0 V vs. Ag/AgCl for 360 s. These deposition times were chosen to achieve similar deposition mass. All three deposition procedures, on average, yield 3 mg cm^{-2} of dendritic copper onto the surface. It is worthy of mention that sodium and lactate ions were highly soluble in water. A rinsing process was conducted after fabrication to ensure the basic samples were free from surface contamination. In addition, the high temperatures and aqueous media of the boiling experiments were sufficient to further remove any possible impurities that may have remained on the surface.

Fig. 2 shows the scanning electron microscopy (SEM) images of copper foams deposited in an acidic solution (pH 0, Fig. 2a-c), and two basic solutions i.e. (pH 10, Fig. 2d-f) and (pH 12, Fig. 2g-i) at different magnifications. All the samples have multi-tier pores and microcavities formed by hydrogen bubbling during electrodeposition. No salt build-up or other residue materials are observed in the SEM images. This is also confirmed by energy-dispersive X-ray spectroscopy (EDX) analysis which shows copper as the major component of the dendritic structures with negligible background signals from carbon and oxygen. Hierarchical copper foams deposited at pH 10 and pH 12 are shown to have

denser and smaller dendrites in micro-leaf structures and smaller gaps between dendrites than those deposited at pH 0. This is consistent with the X-ray diffraction (XRD) analysis reported in [101], where the crystallite size of pH 10 copper foams is shown to be 52 – 57 % smaller than pH 0 copper foams. Fig. 3 shows the XRD measurements of the Cu (111) and Cu(200) peaks of the electrodeposited copper foams at pH 0, pH 10, and pH 12 (Fig. 3a) and the Debye-Scherrer crystallite size along the (111) and (200) orientations based on XRD analysis (Fig. 3b). The broader peaks observed for base deposited copper foams (pH 10, pH 12) indicate that they have smaller crystallite sizes than acid deposited copper foams (pH 0). The differences in crystallite size for the two base-deposited copper foams are smaller than the observed variance between identically-prepared samples (roughly 5 nm).

2.2. Wickability characterization with acoustic emission

Pendant droplet radial wicking tests have been widely used for the wickability characterization of boiling surfaces [20,41,66–68], whereas the recorded signal is limited to optical imaging. The present work uses a custom-built radial wicking setup with synchronized optical and acoustic sensing. As shown in Fig. 4a, a perfluoroalkoxy (PFA) capillary tube with an inner diameter of 0.02" (0.508 mm) is filled with deionized water to form a pendant drop. The textured surface (in this study, copper foam samples) is placed on a translational stage (Newport DS 40-XYZ). A high-speed camera (Phantom VEO 710L) is used to record the liquid level inside the PFA tube at a frame rate of 3,000 fps. During the wicking tests, the translation stage is moved up slowly to initiate the contact of the pendant droplet and the textured surface. The liquid level (H) as a function of time is extracted from the high-speed videos using grayscale-based interface tracking in MATLAB. Existing studies have reported the peak AE frequencies during boiling to be in the range of 1 – 100 kHz [98,99,103,104]. In this work, a MISTRAS R3a AE sensor characterized by a resonant response at 30 kHz was selected for AE sensing in both wicking and boiling tests. While the sensor can be responsive to wider frequency ranges (e.g., 1 kHz – 1 MHz), the accuracy and reliability of the measurements will be significantly lower for signals far from the sensor's resonant frequency. The AE sensor was attached to the translation stage with a magnetic hold down and connected to a data acquisition unit (MISTRAS USB-AE) to record the acoustic signals, including AE count, AE energy, AE hits, and AE amplitudes. The test facility was installed on a Thorlab Nexus optical table (T46V) with passive legs (PTP703) for vibration isolation. A threshold amplitude of 24 dB was set as the baseline which filtered the background acoustic signals in the data acquisition software (MISTRAS AEWIn) during data sampling. The inset of Fig. 4a shows a representative image and acoustic signal (AE amplitude) recorded during a wicking test. A spike in AE amplitude at the onset of wicking is observed, which represents the acoustic features during capillary wicking. This spike can be attributed to the sudden release of energy when the pendant drop is sucked by the wicking surfaces due to capillary forces. The high wicking flux in the initial stage leads to strong stress waves on the surface, which are detected by the AE sensor. As wicking slows down, the AE amplitude also attenuates after the initial spike.

2.3. Pool boiling tests with acoustic emission

Fig. 4b shows the pool boiling test facility, which consists of a closed boiling chamber with a condensing loop for pressure control, a heating element with proper insulation and power supply, a multimodal sensing system including temperature, pressure, and acoustic emission, and the associated data acquisition system. The boiling chamber consists of a polycarbonate ceiling and side walls, and a PEEK base. It has an inner volume of $6.5 \times 6.5 \times 7$ in³ (165.1 mm $\times$ 165.1 mm $\times$ 177.8 mm) and is filled with deionized water to a liquid level of 4 in (101.6 mm) before testing. The ceiling of the boiling chamber is connected to a Graham condenser (Ace glass 5953–106) for degassing and a coiled copper tube

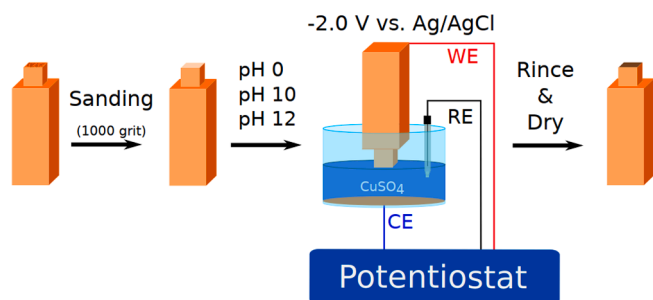


Fig. 1. Schematic diagram of the fabrication process for electrodeposited copper foams on a copper block.

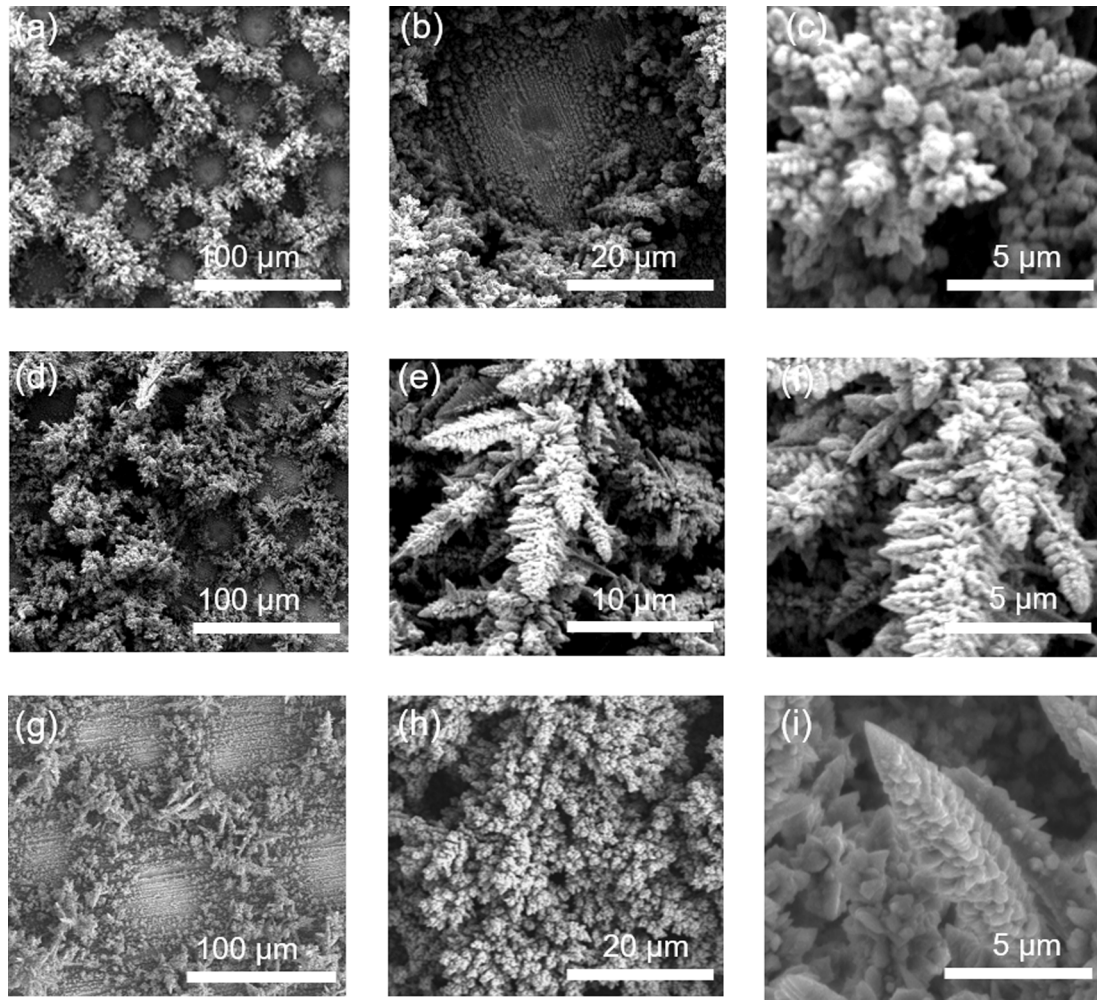


Fig. 2. SEM images at different magnification levels for copper foams fabricated at (a-c) pH 0, (d-f) pH 10, and (g-i) pH 12.

condenser for pressure control, both of which are connected to a chiller (Thermo Scientific Polar ACCEL 500 Low/EA) using a bypass loop. Two compact screw-plug immersion heaters (McMaster-Carr 4668 T54) are installed inside the boiling chamber for degassing and auxiliary heating. A variable-voltage transformer (McMaster-Carr 6994 K17) is used to supply heat to the immersion heaters. The heating element includes a copper block with a boiling surface area of 10 mm × 10 mm, a custom-built PEEK insulation chamber, and nine cartridge heaters (Omega Engineering HDC19102) inserted into the bottom of the copper block. The surface of the copper block is exposed to the liquid pool through the PEEK chamber. The junction between block-PEEK is sealed using RTV (Momentive RTV 106) and epoxy (3 M DP 110) as shown in Fig. 4b and unidirectional heat transmission is ensured. The gap between the copper block and the PEEK chamber is filled with fiberglass and a ceramic block is used at the bottom of the PEEK chamber to support the copper block. The cartridge heaters are connected to a DC power supply (MagnaPower SL200-7.5) for heating power control. Four T-type thermocouples (Omega Engineering TJ36-CPSS-032U-6) are evenly spaced in the copper block for surface temperature and heat flux measurements. A high-accuracy pressure transducer (Omega PX409-030A5V) is used to measure the vapor pressure inside the boiling chamber. Two T-type threaded pool thermocouples (McMaster-Carr 1245N16) are placed inside the boiling chamber to measure liquid and vapor temperatures, respectively. The signals from the thermocouples and pressure transducer are connected to an NI DAQ system (chassis: cDAQ-9178, temperature module: NI 9210, analog input module: NI 9239), and recorded using LabVIEW. A high-speed camera (Phantom VEO 710L) is used to visualize

bubble dynamics during boiling with an LED light source (Advanced Illumination BT200100-WHIIC) for backlighting. The boiling setup uses the same AE sensor, data acquisition unit, and software as the wickability setup reported in Sec. 2.2. After several iterations on the position of the AE sensor for recording the signals, the bottom edge of the boiling chamber that was closer to the heater surface was selected to install the AE sensor. All tests were performed under the same conditions including sensor location, and the analysis is focused on the comparison between the tests using different surface structures. A higher threshold amplitude of 55 dB (compared to 24 dB in wicking tests) was used for the boiling tests to account for the background noises due to heaters and the condensing loop during boiling. The inset of Fig. 4b shows representative AE signals (AE amplitude versus time) during pool boiling which corroborates the strong acoustic signals during boiling phenomena. The temperature, pressure, and high-speed imaging signals are synchronized using hardware connection with BNC cables and Software control in LabVIEW, which reduces the time delay to 5.69 ms [105]. All experimental data reported in this paper are obtained from steady-state pool boiling tests using deionized water. At the beginning of each experiment, water is degassed for half an hour with the Graham condenser open. After degassing, the Graham condenser is turned off and the coiled copper tube condenser is turned on for pressure control. At each heating power, the system reaches a steady state in approximately 15 min and data sampling is performed during this time.

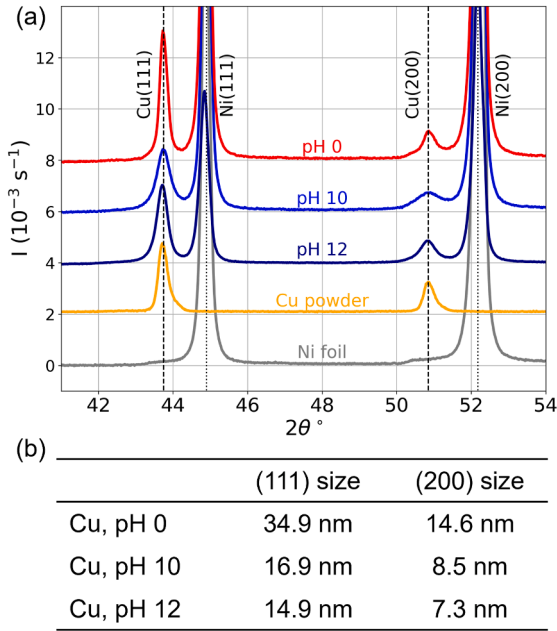


Fig. 3. (a) XRD measurements of the (111) and (200) peaks of the nickel foil substrate, copper powder, and electrodeposited dendritic copper foams at pH 0, pH 10, and pH 12. Ni foil was used as a substrate to study the deposited copper dendrites without being obscured by the copper substrate. (b) Debye-Scherrer crystallite size along the (111) and (200) orientations in electrodeposited dendritic copper foams at pH 0, pH 10, and pH 12.

3. Results and discussion

3.1. Wicking and boiling characterization of copper foams

Fig. 5a shows the wicked volume versus time for both acid-deposited copper foam (pH 0) and base-deposited copper foams (pH 10 & 12), where the symbols and shaded regions represent the mean wicked volume and standard deviations analyzed from six tests per case (two samples per surface, three tests per sample). The wicked volume is determined using the instantaneous liquid level (H) extracted from high-speed imaging following $V_w(t) = \pi d_{\text{tube}}^2 [H_0 - H(t)]/4$, where H_0 is the initial liquid level before wicking and d_{tube} is the inner diameter of the

capillary tube. Despite the relatively large standard deviation due to sample contaminations of the wicking surface, the area of selection for wicking and its non-uniformity, and uncertainties in interface tracking, Fig. 5a shows that the wicked volume of the base deposited copper foams (pH 10 & 12) is higher than that of the acid deposited copper foam (pH 0). Following the convention in [20], the wicking flux is calculated as $\phi_w = (dV_w/dt)|_{t \rightarrow 0}/A_w$, where A_w is the wetted area measured from the optical image ($\sim 0.48 \text{ mm}^2$ in the present study). The wicking fluxes and peak AE amplitudes during wicking on electrodeposited copper foams at pH 0, pH 10, and pH 12 are listed in Table 1. Since polished copper is not a wicking surface, the wicking flux and AE-wicking entries are kept empty in Table 1.

The calculated wicking fluxes for pH 0, 10, and 12 copper foams are also plotted in Fig. 5b. It is observed that the wicking flux increases with the pH, with a mean wicking flux of 30.7 mm/s for pH 0, 41.8 mm/s for pH 10, and 47.3 mm/s for pH 12. To understand the role of surface structures in enhanced wicking, Fig. 6 plots the mean wicking flux as a function of the crystallite size based on Cu(111) and Cu(200) surfaces for copper foams at pH 0, pH 10, and pH 12. It is observed that the wicking flux increases as the crystallite size decreases following a nearly inverse relationship. This observation can be attributed to the higher capillary pressure at smaller structure sizes. It is known that many physical forces play a role in different stages of capillary wicking, including surface tension, viscous force, inertia, and gravity. Existing wicking models are predominantly focused on the viscous limit, where capillary pressure is balanced by viscous resistance between the liquid and solid. As such, the wicking metrics defined by these studies normally include the product of the capillary pressure and the permeability (the inverse of the permeability represents the strength of viscous resistance) [59,60,67]. Nevertheless, the viscous limit fails to describe the upper limit of wicking fluxes since it leads to an unbounded singularity in the early stage of capillary wicking when the wicking length vanishes. Compared to viscous resistance, inertia plays a more critical role in early-stage wicking [106]. In the inertia limit, the wicking flux is dictated by the capillary pressure and the influence of viscous resistance is negligible. The smaller dendritic features present in copper foams at pH 10 and pH 12, particularly the average smaller gaps between dendrites in these foams, can lead to increased capillary pressure and thus stronger surface wickability [107].

The pool boiling results of the polished copper surfaces and the acid and base-deposited copper foams are presented in Fig. 7. Fig. 7a shows the heat flux (q'') as a function of the wall superheat (ΔT), and Fig. 7b

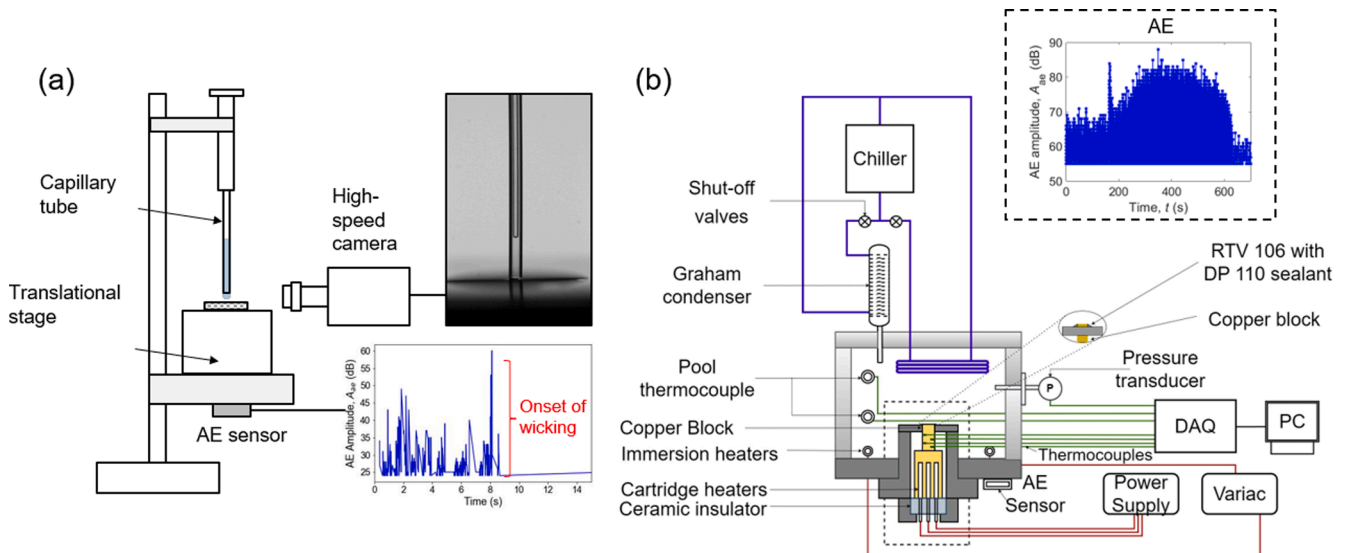


Fig. 4. Schematic diagrams of (a) the surface wickability testing facility with the inset showing a representative optical image and acoustic signals and (b) the pool boiling test facility with the inset showing representative acoustic signals.

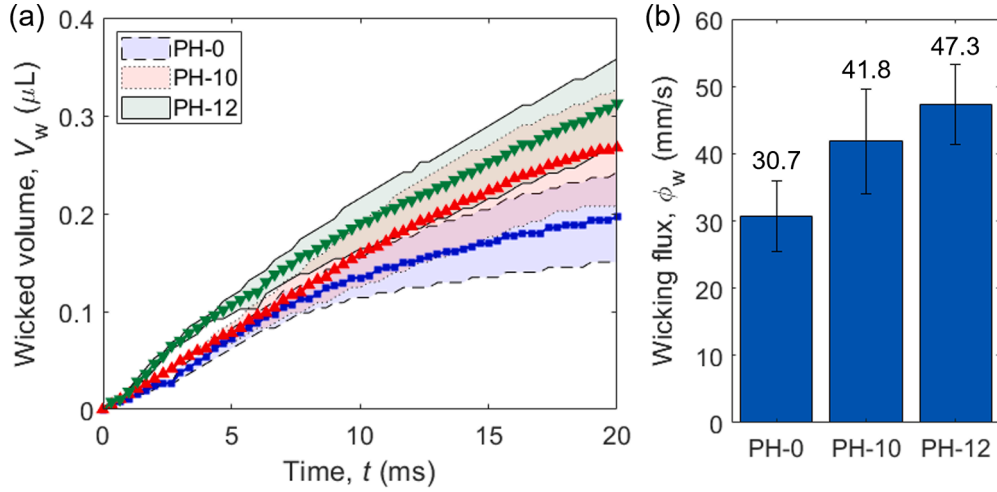


Fig. 5. Wickability of copper foams showing (a) wicking volume versus time and (b) wicking flux for acid-deposited copper foams (pH 0) and base-deposited copper foams (pH 10 and pH 12). The symbols and shaded areas represent the mean and standard deviation, respectively.

Table 1

Summary of the wicking flux, critical heat flux, and heat transfer coefficient of polished copper and copper foam structures used in the present work.

Heater surface	Wicking flux (mm/s)	AE-wicking (dB)	CHF (W/cm ²)	Max. HTC (W/(cm ² K))	AE-CHF (dB)
Polished copper	–	–	102.6	6.4	58.2
Copper foam at pH 0	30.7	65.5	183.9	7.8	62.0
Copper foam at pH 10	41.8	39.2	218.6	10.7	59.6
Copper foam at pH 12	47.3	38.2	202.1	9.3	59.8

copper foams are summarized in Table 1. The CHF value for the polished copper surface is 102.6 W/cm², which is close to Zuber's limit of 110 W/cm². The maximum HTC for a polished copper surface is $h = 6.4$ W/cm² K at a wall superheat ($\Delta T = 15.83$ °C). Both acid-deposited copper foams (pH 0) and base-deposited copper foams (pH 10 & 12) yield boiling CHF well beyond the Zuber's limit. Particularly, the pH 10 copper foam leads to the highest CHF among the foams at 218.6 W/cm², which is 113 % larger than polished copper. The copper foams generally have higher surface superheat than polished copper at the same heat flux, owing to the higher nucleation barrier due to the superhydrophilicity of the foam structures. As a result, the HTC of copper foams is lower than that of polished copper at the same heat flux, as shown in Fig. 7b. Nevertheless, all the hierarchical copper foams still lead to higher maximum HTCs than polished copper surfaces because the ratio of heat flux increment against the increase in temperature superheat was larger. As such HTCs of 7.8 W/cm² K, 10.7 W/cm² K, and 9.3 W/cm² K were obtained for pH 0, pH 10, and pH 12 copper foams, respectively. The Rohsenow correlation is used to benchmark the boiling curve on polished surfaces in Fig. 7a&b using two different coefficients of liquid-surface combinations, i.e., $C_{sf} = 0.0128$ [108] and $C_{sf} = 0.0107$ [109]. The boiling curve of the polished Cu surface is shown to agree well with the Rohsenow correlation with $C_{sf} = 0.0107$ suggesting a new liquid surface combination for water-sandpaper polished copper surface. Boiling inversion is observed in all copper foam cases, represented with kinks in the boiling curves roughly in the heat flux range of 80–110 W/(cm²K). These kinks indicate the bubble dynamics have gone through a transition, which will be discussed in Sec. 3.3.

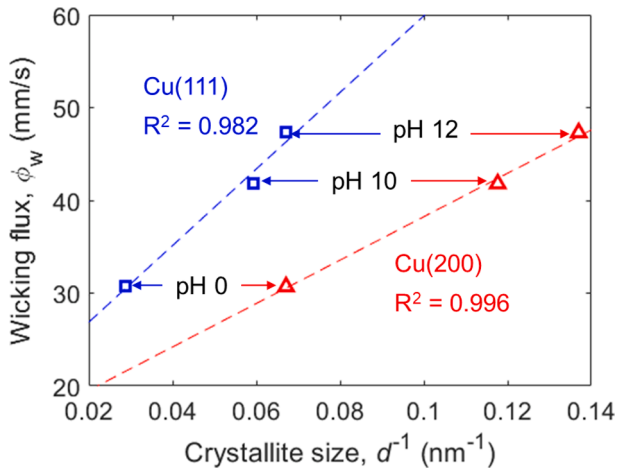


Fig. 6. Wicking flux as a function of the inverse of the Debye-Scherrer crystallite size along the (111) and (200) orientations for electrodeposited copper foams at pH 0, pH 10, and pH 12.

shows the heat transfer coefficient (h) as a function of the heat flux. The inset of Fig. 7b illustrates the wicking-based mechanism of heat transfer enhancement during boiling on copper foams, where liquid from the pool is imbibed to the liquid layer underneath the bubble via capillary wicking to delay dryout. The CHF and HTC of polished copper and

3.2. Acoustic emission characteristics of wicking and boiling

The mechanism of structure-induced boiling CHF enhancement is probed using acoustic sensing in both wicking and boiling tests. Fig. 8 shows AE signals captured during the CHF event of water boiling on pH 0 copper foam, with Fig. 8a showing the AE amplitude as a function of time throughout the test, and Fig. 8b and 8c showing representative waveforms in the nucleate boiling regime and CHF, respectively. The black dashed lines represent the preset threshold voltage (V_{th}) corresponding in decibels in a waveform which is 57.7 mV equivalent to 55 dB set in the boiling tests. When the measured voltage goes beyond the defined threshold, a hit from the waveform is recognized. The number of hits, the peak voltage, the time from the first hit to the peak voltage (defined as rise), and the duration of the waveform are the commonly used AE parameters to analyze the signals. The peak decibel or voltage of a waveform is named the AE amplitude and is plotted in Fig. 8a. The

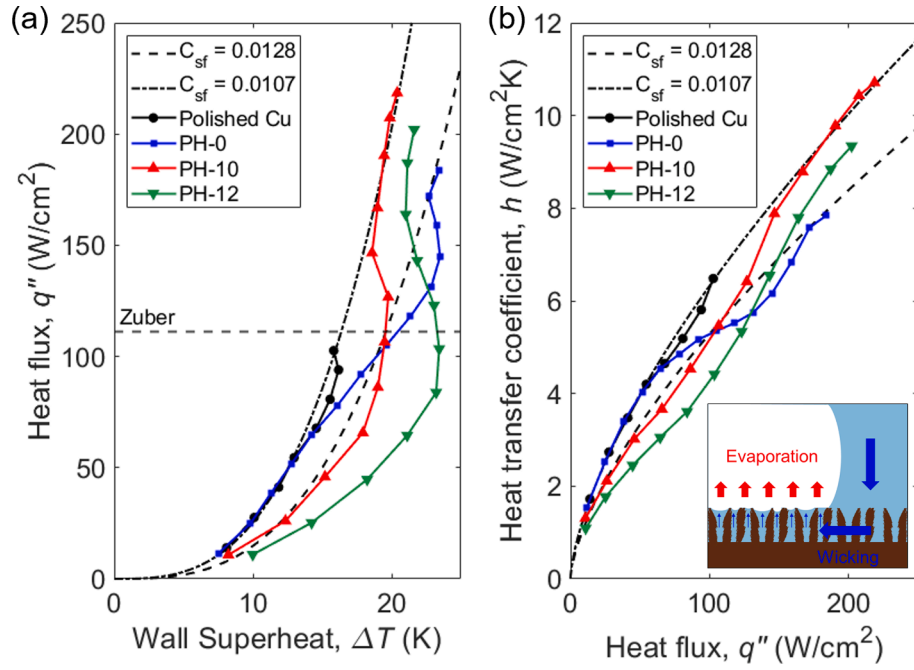


Fig. 7. Pool boiling heat transfer performance of the polished copper surface and copper foams deposited at pH 0, 10, and 12 showing (a) boiling heat flux as a function of the wall superheat with the Rohsenow correlation at different liquid surface combinations (C_{sf}) for water on polished copper and (b) Heat transfer coefficient as a function of the boiling heat flux with the Rohsenow correlation. The inset illustrates the wicking-based boiling enhancement mechanism.

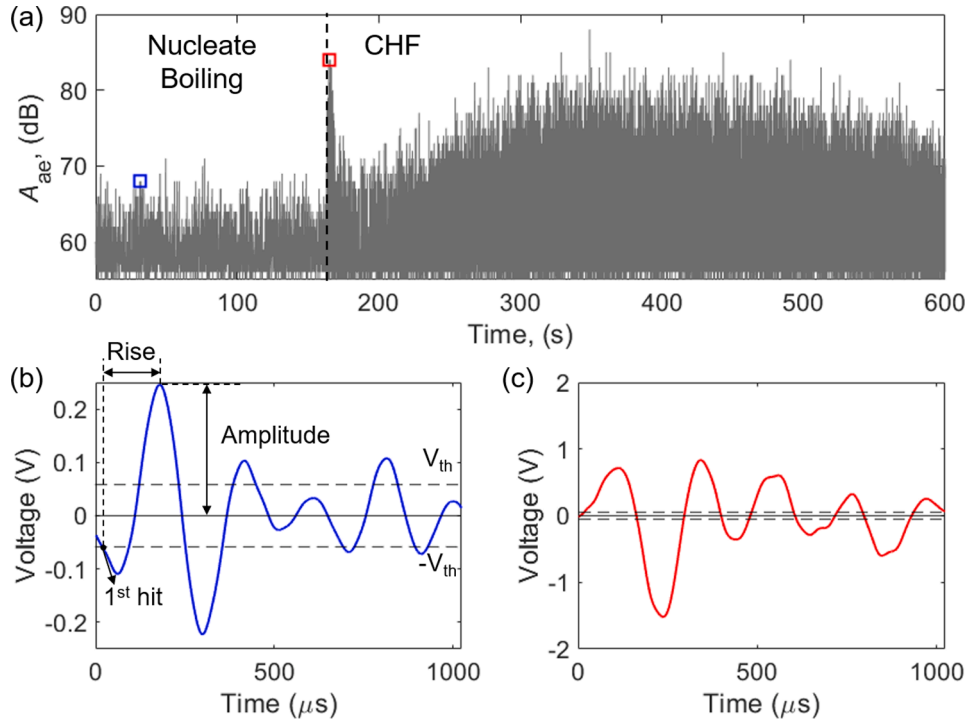


Fig. 8. Representative AE signals during boiling showing (a) AE amplitude versus time and (b-c) representative waveforms during (b) nucleate boiling and (c) critical heat flux on pH 0 copper foam.

time interval between the first hit and the peak is called the rise. As shown in Fig. 8b and 8c, the AE amplitude at CHF (1.52 V in terms of voltage) is much higher than pre-CHF (0.24 V). The sound levels in decibels follow a logarithm relationship with the voltage and are 84 dB at CHF and 68 dB at the moment represented by Fig. 8b. The rise at CHF (237 μ s) is also higher than that in the nucleate boiling (179 μ s at the moment represented by Fig. 8b). It is evident from Fig. 8 that AE

amplitude is very sensitive to the boiling regime and can thus be used to represent boiling regimes and the CHF conditions.

Fig. 9 shows the synchronized temperature, heat flux, and AE amplitude data versus time for water boiling on (a-b) polished copper and (c-d) electrodeposited copper foam at pH 0. The four temperature readings, i.e., T_1 , T_2 , T_3 , and T_4 , are recorded from the thermocouples evenly spaced in the copper block with light to dark gray representing

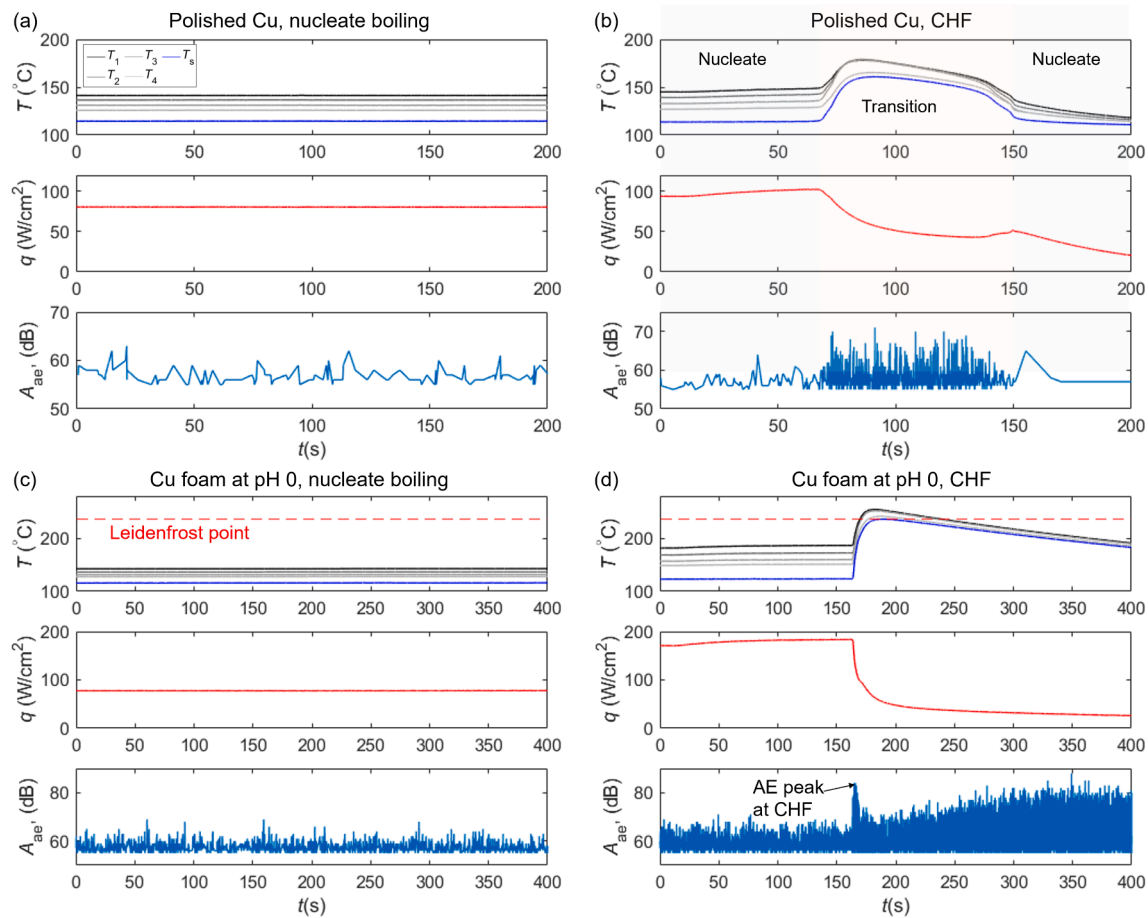


Fig. 9. Synchronized temperature, heat flux, and AE amplitude for water boiling on (a-b) polished copper showing (a) representative nucleate boiling at a heat flux of 80.7 W/cm^2 and (b) the CHF condition and (c-d) electrodeposited copper foam at pH 0 showing (c) representative nucleate boiling at a heat flux of 78.0 W/cm^2 and (d) the CHF condition.

the direction from the boiling surface to the cartridge heaters, and T_s represents the extrapolated boiling surface temperature. The AE amplitude is recorded by the AE sensor attached to the boiling chamber. For water boiling on polished copper, Fig. 9a and 9b show a representative steady-state nucleate boiling condition ($q'' = 80.7 \text{ W/cm}^2$) and the CHF condition, respectively. As shown in Fig. 9a, during steady-state nucleate boiling, the AE amplitude has some fluctuations due to background noises but mostly fluctuates around a constant value. Fig. 9b shows the synchronized multimodal signals under a heat load of 120 W , which triggers CHF at $t \approx 68 \text{ s}$. Upon the incipience of CHF, the temperature ramps up rapidly due to the additional thermal resistance induced by the vapor layer blanketing the heater surface. To avoid overheating-induced device failures, the heaters were turned off immediately after the temperature spike was observed during the boiling test (with a time gap of $\sim 3 \text{ s}$). As a result, temperatures started to decrease after the initial rapid growth. It is noted that water remained in the transition boiling regime for an extended period after the heater was turned off, and returned to nucleate boiling at $t \approx 149 \text{ s}$. The discussions of this extended period are detailed in the authors' other paper [110] and will not be expanded here. Fig. 9b shows higher AE amplitudes in the transition boiling regime, which is consistent with existing studies [93,96,111]. For nucleate boiling of water on copper foam at pH 0, the behavior of the temperature, heat flux, and AE amplitude are very similar to those on the polished copper surface (Fig. 9c). However, the AE signals at the CHF condition are quite different between copper foams and polished copper. As shown in Fig. 9d, a spike in the AE amplitude is observed when CHF is triggered during water boiling on copper foam at pH 0, represented by a sudden increase in AE amplitude

followed by a subsequent sudden decrease. When CHF is triggered, the vigorous bubble dynamics and rapid growth of the vapor layer lead to higher acoustic emissions, represented by the sudden increase in the AE amplitude. The rapidly growing vapor layer covers the entire heater surface very soon and mitigates the disturbances of the liquid-vapor interfaces, resulting in very slow ebullition and a sudden decrease in the AE amplitude. As demonstrated by Seo and Bang [96] and Sinha et al. [93], the AE amplitude in the film boiling regime is very low due to the stable vapor film (referred to as "silent regime" by Sinha et al.). In the present work, the heater was turned off very soon after CHF was observed (within 3 s). The loss of power makes the vapor layer metastable, leading to stronger interface disturbances, represented by an increasing AE amplitude. This explains the observation of a spike followed by a gradual increase in the AE amplitude in Fig. 9d. This trend is consistent with the AE signals observed by Sinha et al. at the CHF condition [93,112].

It is noted that this type of AE amplitude spikes is observed at CHF during all the boiling tests on copper foams (including pH 0, pH 10, and pH 12), whereas no such spikes are observed during boiling tests using polished copper, as confirmed by multiple independent tests. This difference can be attributed to the higher CHF and surface superheats during boiling on copper foams. For example, for water boiling on electrodeposited copper at pH 0, the heater surface temperature was 123.4°C when CHF was triggered and reached 236.0°C before it started to decrease. This peak temperature is very close to the Leidenfrost point of water on copper surfaces (237°C , as reported in [113]), and is thus sufficiently high to maintain a stable vapor film to blanket the surface. On the contrary, for the polished copper case, the surface temperature

was at 114.4 °C when CHF was triggered, and only reached up to 160.9 °C, which is too low to maintain a stable vapor film.

To understand the governing mechanisms of structure-induced CHF enhancement, the acoustic signatures of copper foam-enhanced boiling heat transfer are compared to the acoustic signatures of wicking represented in Fig. 10. Fig. 10a shows the average AE amplitude at the incipience of wicking for pH 0, pH 10, and pH 12 copper foams from the wicking tests. The acid-deposited copper foam (pH 0) has an AE amplitude of 65.5 dB, higher than that of the two base-deposited copper foams including 39.2 dB for pH 10 copper foam and 38.2 dB for pH 12 copper foam. The difference in AE amplitudes between pH 10 and pH 12 copper foams is negligible compared to the experimental uncertainties. Since the basic deposited copper foams (pH 10 & 12) have higher wicking fluxes than the acid-deposited copper foam (pH 0) as shown in Figs. 5 and 6, the AE amplitude is inferred to have a negative correlation with the wicking flux. This peculiar observation is attributed to the signal measurement mechanism of the AE sensor and the swift-wicking process. We suspect the inverse relation is due to the rapidity of wicking processes occurring at a thousandth of a second which has waning effects on its acoustic signatures. As such, it is expected that the AE sensor records minimal amplitudes at high-wicking surfaces. To compare the acoustic signatures from wicking and boiling tests, the boiling acoustic signatures are plotted in Fig. 10b (nucleate boiling) and Fig. 10c (CHF) by running the analysis illustrated in Fig. 9 for all boiling tests. As shown in Fig. 10b, the AE amplitude of the acid-deposited copper foam (pH 0) is shown to be higher than the base-deposited copper foams (pH 10 & 12) throughout the nucleate boiling regime except for the low-heat-flux region that has large fluctuations. Despite the AE amplitudes at CHF being higher than nucleate boiling due to the vapor layer formation, the same trend is observed at CHF (Fig. 10c). The AE amplitude of the acid-deposited copper foam (pH 0) is 62.0 dB, larger than the base deposited copper foams, including 59.6 dB for pH 10 copper foam and 59.8 dB for pH 12 copper foam. Similar to the wicking acoustic signatures, the difference in the boiling AE amplitude between pH 10 and pH 12 is negligible. By comparing Fig. 10a and Fig. 10c, a consistent trend in AE amplitude is observed between wicking and boiling tests: both the wicking flux and boiling CHF decrease with the AE amplitude. This observation provides evidence for the critical role that capillary wicking plays in structure-induced CHF enhancement from a new perspective (acoustic emission). It is also shown in Fig. 10b that in the nucleate boiling regime, the AE amplitude first decreases with the heat flux (roughly, $q'' < 100 \text{ W/cm}^2$) and then increases with the heat flux. This observation is believed to be correlated to the bubble dynamics during boiling and will be explained in Sec 3.3 with the analyzed bubble size and count.

3.3. Bubble dynamics and vapor fraction analysis

The bubble dynamics during water boiling on polished copper and copper foam surfaces are analyzed using high-speed videos captured during boiling tests. Fig. 11 shows representative bubble images from boiling tests on the polished copper surface and pH 0, pH 10, and pH 12 copper foams at different heat fluxes. The heat flux values correspond to the boiling curves in Fig. 7 before CHF. Irrespective of the different wicking properties for polished copper and copper foam samples as shown in Table 1, a common trend can be observed in all cases that discrete bubbles are observed at low heat fluxes ($<30 \text{ W/cm}^2$) and bubbles tend to coalesce and form a mushroom-shaped cloud as the heat flux increases. It can also be observed that bubble coalescences majorly take place near the heater surface to feed vapor to the giant cloud above. As heat flux further increases, the edge of the mushroom-shaped cloud tends to move closer toward the heater surface. To quantitatively analyze bubble dynamics, Fig. 12a and 12b show the maximum bubble diameter ($D_{\max}$) and bubble count (N) versus heat flux, respectively, extracted from high-speed videos using ImageJ [114]. The error bars are generated from averaging results from multiple frames per heat flux. Since the bubble size varies in an ebullition cycle, the maximum bubble diameter of each ebullition cycle is used. As shown in Fig. 12a, the maximum bubble diameter increases with the heat flux before 80 W/cm^2 due to more frequent bubble coalescences, which is also supported by the rapid decrease of the bubble count with heat flux shown in Fig. 12b. For heat fluxes beyond 80 W/cm^2 , the increase/decrease in $D_{\max}/N$ with heat flux becomes much slower as coalescence has become frequent enough to absorb newly nucleated bubbles into the vapor cloud. In this regime, the increase in the heat flux is characterized by the larger vapor cloud. This transition of bubble dynamics corresponds to the kinks observed in the boiling curves. The analyzed bubble size and bubble count can also be used to explain the observed relationship between the AE amplitude and heat flux in the nucleate boiling regime (Fig. 10b). Assuming the shape of the bubbles to be a spherical cap, the force that a bubble exerts on the surface is $F = \sigma \pi D_{\text{bubble}} \sin \theta \tan \theta$, where D_{bubble} is the bubble diameter and θ is the contact angle [115]. The overall stress that bubbles exert on the heater surface is thus $\tau = N \sigma \pi D_{\text{bubble}} \sin \theta \tan \theta / A_{\text{heater}}$, where N is the bubble count and A_{heater} is the heater footprint. Based on this analysis, the bubble-induced stress on the heater surface is positively correlated with both D_{bubble} and N following $\tau \sim D_{\text{bubble}} / A_{\text{heater}}$. Existing experimental studies have reported the positive relevance between the AE amplitude and bubble size [116–118]. For example, bubble incipient tests show an average 1.1 dB increase in the AE amplitude when the bubble diameter doubles [118]. In the early nucleate boiling stage ($q'' < 100 \text{ W/cm}^2$), the rapid bubble size increase with heat flux leads to stronger AE signals per bubble.

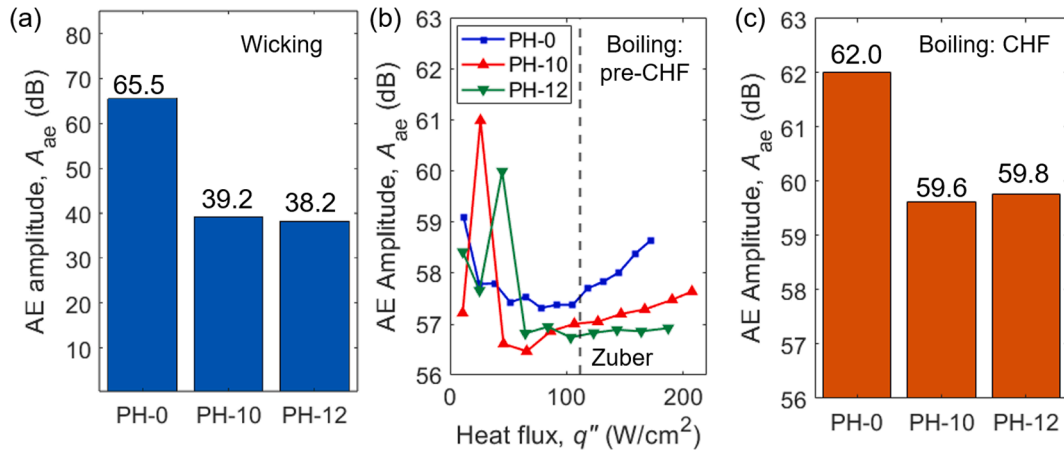


Fig. 10. Acoustic emission characteristics in wicking and boiling tests. (a) Average AE amplitude at the onset of wicking in the wicking tests. (b) Average AE amplitude during pool boiling in the nucleate boiling regime. (c) Average AE amplitude during CHF.



Fig. 11. Representative boiling images at different heat fluxes for polished copper and copper foams deposited at pH 0, pH 10, and pH 12. The scale bar represents 10 mm.

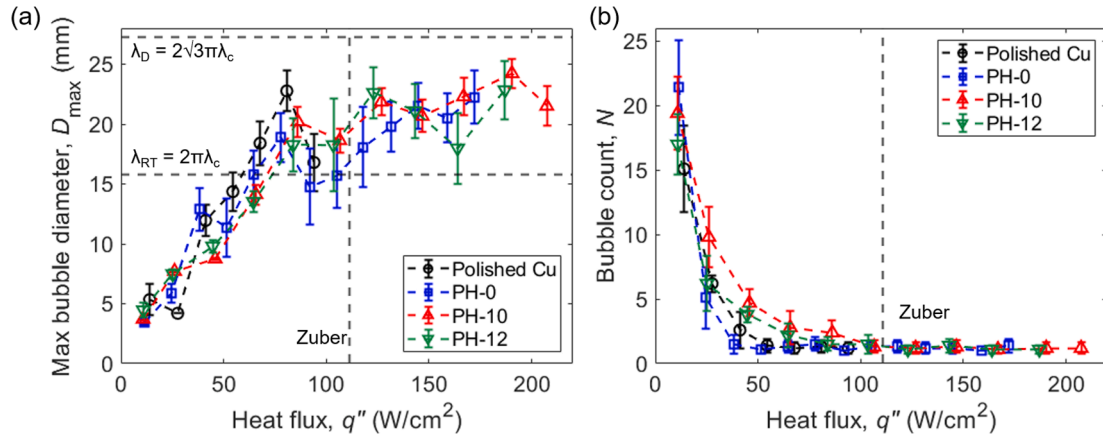


Fig. 12. Bubble dynamics analysis during pool boiling showing (a) the maximum bubble diameter and (b) bubble count at different heat fluxes for water boiling on polished copper and copper foams deposited at pH 0, pH 10, and pH 12.

Nevertheless, the bubble count reduces more significantly with heat flux in this stage. Using a power law fitting of the data in Fig. 12, it was found that, in the range of $q'' < 100 \text{ W/cm}^2$, the maximum bubble diameter

and bubble count scale with the heat flux following $D_{\max} \sim q''^{0.74}$ and $N \sim q''^{-1.27}$, respectively. As such, the stress on the heater surface is expected to scale with the heat flux as $\tau \sim q''^{-0.53}$. Therefore, the AE

amplitude decreases with the heat flux in this stage. For $q'' > 100 \text{ W/cm}^2$, the bubble count is nearly saturated at $N = 1$ (giant mushroom-shaped cloud) while the bubble diameter still slowly increases with the heat flux following $D_{\max} \sim q''^{0.35}$. Thus, the AE amplitude starts to increase with the heat flux in the stage, which is consistent with Negi et al.'s observation of increasing AE amplitude with heat flux in the range of $q''/q''_{\text{CHF}} = 0.4 - 0.95$ [97].

In hydrodynamic instability-based CHF theories [5,6], CHF is triggered by the Helmholtz instability at the interface of vapor columns leaving the heating surface and the critical wavelength λ_C of Rayleigh-Taylor instability is used to estimate the distance between neighboring vapor columns in a large boiling surface. The generated bubble perturbations with wavelengths larger than critical wavelength lead to the instability of the vapor-liquid interface. As such, two characteristic wavelengths of Rayleigh-Taylor instabilities are plotted in Fig. 12a as reference length scales to correlate our maximum bubble departure diameter in the pool boiling test with an available theoretical explanation. The critical wavelength that initiates Rayleigh-Taylor instability is $\lambda_{\text{RT}} = 2\pi \lambda_C$, where λ_C is the capillary length that can be determined with liquid surface tension (σ), the gravity acceleration (g), and the difference in liquid density (ρ_l) and vapor density (ρ_v) using $\lambda_C = \sqrt{\sigma/(\rho_l - \rho_v)g}$. The wavelength that corresponds to the maximum disturbance, aka the most dangerous wavelength, is $\lambda_D = 2\pi\sqrt{3}\lambda_C$. For saturated water at 1 atm, the capillary length is 2.51 mm, leading to a λ_{RT} of 15.7 mm and λ_D of 27.3 mm. It is observed in Fig. 12a that $D_{\max}$ falls between λ_{RT} and λ_D for all cases when the heat flux is beyond Zuber's limit. It can also be observed that the differences in $D_{\max}$ and N between different boiling surfaces (polished copper, copper foams deposited at varying pH) at the same heat flux are not pronounced (within the error bars).

The analysis of $D_{\max}$ and N is a lumped steady-state analysis that does not account for the dynamic oscillations during steady states. To probe the dynamic behavior of the bubbles, the area fraction of the vapor phase is analyzed at every frame of the high-speed videos using the approach detailed in [119]. By tracking the interface motion using consecutive boiling images, a vapor mask is extracted at each frame to determine the vapor fraction (see the inset of Fig. 13a for a representative bubble image and the corresponding vapor mask). Since this analysis is affected by the magnification of optical imaging, the calculated vapor fraction is scaled using the scale bar in each case. Fig. 13 shows the representative vapor fraction analysis as a function of time where Fig. 13a is steady-state boiling of water on the pH 12 copper foam at $q'' = 103.4 \text{ W/cm}^2$ extracted from the high-speed video. Each of the steady-state boiling experiments conducted over various copper samples has different heat flux stages starting from early bubble nucleation, and bubble coalescence to vapor blanketing over the boiling surfaces. During these boiling stages, the vapor fraction changes with the varying heat fluxes as bubble nucleation density increases followed by the formation of larger bubbles and vapor slugs at larger heat fluxes. Also, in each of

these individual steady-state heat flux stages, the vapor fraction oscillates over time as bubbles are nucleated, coalesced, and departed from the heater surface. An envelope analysis is performed with ϕ_{up} and ϕ_{low} representing the upper limit and lower limit of the envelope of the vapor fraction data and ϕ_m the mean line calculated using $\phi_m = (\phi_{\text{up}} + \phi_{\text{low}})/2$. Spectrum analysis is performed using the Fast Fourier Transform of the scaled vapor fraction data, $\phi(t) - \phi_m(t)$, and the results are plotted in Fig. 13b. It is observed that the key features of the signals are in the range of 8 Hz to 12 Hz. The power and the dominant frequency of vapor fraction are analyzed from the spectrum with a threshold amplitude imposed to filter out high-frequency noises for all boiling cases (polished copper & copper foams deposited at pH 0, 10, and 12). The dominant frequency of the vapor fraction represents the most critical time scale at which the bubbles or the mushroom-shaped bubble cloud behaves, and the power represents the variance of the vapor fraction. The results are shown in Fig. 14. The power levels of different boiling frequencies obtained from vapor fraction analysis are shown to increase monotonically with the heat flux. The data in Fig. 14a also reveal kinks at $q'' \sim 80 \text{ W/cm}^2$ similar to the boiling curve (Fig. 7a) and maximum bubble diameter (Fig. 12a), signifying the transition of the bubble coalescence regime to the increasing vapor cloud regime. The dominant frequency of the vapor fraction does not show a strong dependence on the heat flux. Particularly, for heat fluxes beyond Zuber's limit, the dominant frequency converges for all cases and fluctuates around 10 Hz. This dominant frequency represents the oscillation of the mushroom-shaped cloud. It is in the same order of magnitude but is generally smaller than the reported departure frequency of individual bubbles at low heat fluxes [41,78,79,81]. Based on the analysis of maximum bubble diameter and bubble count depicting larger standard deviations, as well as supported with vapor fraction data for different surfaces with their varying wickability, it does not show an obvious effect of wicking on the bubble dynamics while comparing equivalent heat flux for each of the boiling surfaces. It is thus inferred that the structure-induced CHF enhancement is not directly related to the interface of the large-scale liquid-vapor dynamics. Combining this observation with the acoustic signals reported in Sec. 3.2 shows a similar trend of AE signature during wicking and boiling, the present work supports that the structure-induced CHF enhancement lies in the stronger capillary wicking and the effective liquid entrainment to the growing bubble rather than the apparent bubble dynamics.

4. Conclusion

In this paper, electrodeposition with pH control is used to fabricate copper foam structures for boiling enhancement. Copper foams deposited at higher pH are shown to yield stronger wickability owing to the smaller dendrites and gaps between dendrites. Pool boiling experiments are performed to characterize the thermal performance of the fabricated foams and show higher CHF on base-deposited copper foams (pH 10 &

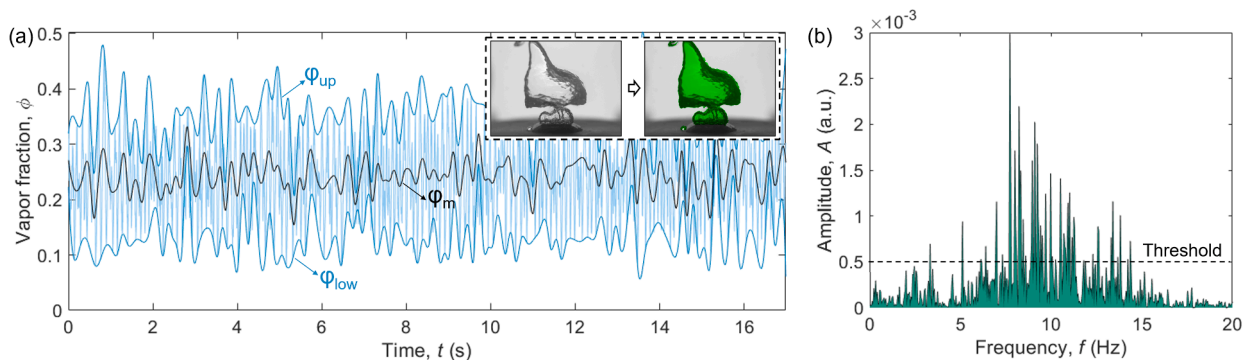


Fig. 13. (a) Vapor fraction as a function of time and (b) power spectrum calculated from vapor fraction analysis for water boiling on pH 12 copper foam at $q'' = 103.4 \text{ W/cm}^2$.

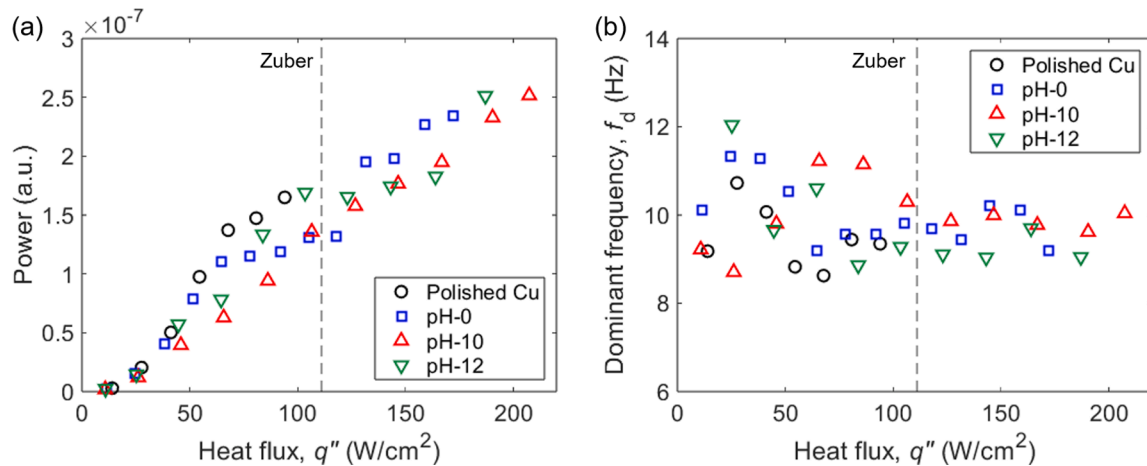


Fig. 14. Spectral analysis of vapor fraction during boiling showing (a) power of the vapor fraction spectral as a function of heat flux and (b) dominant frequency as a function of heat flux.

12) than on acid-deposited foams (pH 0). The mechanism of the foam-induced CHF enhancement is probed using optical imaging and acoustic emissions. The acoustic signatures are investigated during steady-state boiling and an AE parameter in terms of amplitude is used to characterize both the wicking tests and the boiling tests using nano-engineered surfaces. The results have shown higher AE amplitude in the transition boiling regime, which is consistent with existing studies. From the AE amplitude analysis, pH 10 and pH 12 copper foams with larger wicking properties are shown to have a lower AE amplitude than pH 0 copper foams. The similar trends observed in the AE signals during wicking tests and pool boiling tests indicate wicking to be the key mechanism for CHF enhancement. Bubble dynamics are characterized using steady-state parameters including maximum bubble diameter and bubble count and dynamic oscillations of vapor fraction extracted from high-speed videos captured during boiling tests. No obvious differences in bubble dynamics are observed between the tested boiling surfaces (polished copper, pH 0, 10, and 12 copper foams) at similar heat fluxes. It is thus inferred that the structure-induced CHF enhancement is not directly related to the interface of the large-scale liquid–vapor dynamics. During the nucleate boiling regime, the AE amplitude is shown to first decrease with the heat flux rapidly but then increase with the heat flux gradually. This phenomenon is owed to the interplay between bubble size and bubble count with heat flux and their role in the stress field that bubbles exert on the heater surface.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work was supported in part by the Chancellor's Fund for Innovation and Collaboration at the University of Arkansas and an Arkansas NASA EPSCoR Research Infrastructure Development (RID) award. The authors would like to thank Professor Nenad Miljkovic and Dr. Jiaqi Li of the University of Illinois at Urbana-Champaign for their valuable discussion about this work.

References

- [1] G. Chen, M. Jia, S. Zhang, Y. Tang, Z. Wan, Pool boiling enhancement of novel interconnected microchannels with reentrant cavities for high-power electronics cooling, *Int. J. Heat Mass Transf.* 156 (2020), 119836, <https://doi.org/10.1016/j.ijheatmasstransfer.2020.119836>.
- [2] S. Fan, F. Duan, A review of two-phase submerged boiling in thermal management of electronic cooling, *Int. J. Heat Mass Transf.* 150 (2020), 119324, <https://doi.org/10.1016/j.ijheatmasstransfer.2020.119324>.
- [3] R. Hernandez, R.J. Armstrong, S. Bin Seo, C.P. Folsom, C.B. Jensen, N.R. Brown, Sensitivity analysis of in-pile CHF experiments in the TREAT facility: characterization of impacts of fuel system thermal properties, *Ann. Nucl. Energy* 165 (2022), 108645, <https://doi.org/10.1016/j.anucene.2021.108645>.
- [4] C. Wang, Y. Zhuang, Y. Dong, L. Zhang, L. Liu, J. Du, Design and control analysis of the side-stream extractive distillation column with low concentration intermediate-boiling entrainer, *Chem. Eng. Sci.* 247 (2022), 116915, <https://doi.org/10.1016/j.ces.2021.116915>.
- [5] N. Zuber, Hydrodynamic Aspects Of Boiling Heat Transfer (Thesis), 1959, doi: 10.2172/4175511.
- [6] V.K. Dhir, S.P. Liaw, Framework for a unified model for nucleate and transition pool boiling, *J. Heat Transfer* 111 (3) (1989) 739–746, <https://doi.org/10.1115/1.3250745>.
- [7] S.G. Kandlikar, A theoretical model to predict pool boiling CHF incorporating effects of contact angle and orientation, *J. Heat Transfer* 123 (6) (2001) 1071–1079, <https://doi.org/10.1115/1.1409265>.
- [8] C.K. Guan, J.F. Klausner, R. Mei, A new mechanistic model for pool boiling CHF on horizontal surfaces, *Int. J. Heat Mass Transf.* 54 (17–18) (2011) 3960–3969, <https://doi.org/10.1016/j.ijheatmasstransfer.2011.04.029>.
- [9] Y. Haramura, Y. Katto, A new hydrodynamic model of critical heat flux, applicable widely to both pool and forced convection boiling on submerged bodies in saturated liquids, *Int. J. Heat Mass Transf.* 26 (3) (1983) 389–399, [https://doi.org/10.1016/0017-9310\(83\)90043-1](https://doi.org/10.1016/0017-9310(83)90043-1).
- [10] L. Zhang, J.H. Seong, M. Bucci, Percolative scale-free behavior in the boiling crisis, *Phys. Rev. Lett.* 122 (13) (2019), 134501, <https://doi.org/10.1103/PhysRevLett.122.134501>.
- [11] L. Zhang, S. Gong, Z. Lu, P. Cheng, E.N. Wang, Boiling crisis due to bubble interactions, *Int. J. Heat Mass Transf.* 182 (2022), <https://doi.org/10.1016/j.ijheatmasstransfer.2021.121904>.
- [12] M.S. Kamel, F. Lezsovit, A.M. Hussein, O. Mahian, S. Wongwises, Latest developments in boiling critical heat flux using nanofluids: a concise review, *Int. Commun. Heat Mass Transfer* 98 (September) (2018) 59–66, <https://doi.org/10.1016/j.icheatmasstransfer.2018.08.009>.
- [13] H. Kim, Enhancement of critical heat flux in nucleate boiling of nanofluids: a state-of-art review, *Nanoscale Res. Lett.* 6 (2011) 1–18, <https://doi.org/10.1186/1556-276X-6-415>.
- [14] A. Pare, S. Kumar Ghosh, The chronological study on parametric evolution of pool boiling with nanofluids: an experimental review, *Therm. Sci. Eng. Progr.* 34 (2022), 101420, <https://doi.org/10.1016/j.tsep.2022.101420>.
- [15] S.H. Kim, G.C. Lee, J.Y. Kang, K. Moriyama, H.S. Park, M.H. Kim, Heat flux partitioning analysis of pool boiling on micro structured surface using infrared visualization, *Int. J. Heat Mass Transf.* 102 (2016) 756–765, <https://doi.org/10.1016/j.ijheatmasstransfer.2016.06.040>.
- [16] S.H. Kim, G.C. Lee, J.Y. Kang, K. Moriyama, M.H. Kim, H.S. Park, Boiling heat transfer and critical heat flux evaluation of the pool boiling on micro structured surface, *Int. J. Heat Mass Transf.* 91 (2015) 1140–1147, <https://doi.org/10.1016/j.ijheatmasstransfer.2015.07.120>.
- [17] B. S. Kim, H. Lee, S. Shin, G. Choi, H. H. Cho, Interfacial wicking dynamics and its impact on critical heat flux of boiling heat transfer, *Appl Phys Lett*, vol. 105, no. 19, 2014, doi: 10.1063/1.4901569.

- [18] H. Kim, Y. Park, H. Kim, C. Lee, D.W. Jerng, D.E. Kim, Critical heat flux enhancement by single-layered metal wire mesh with micro and nano-sized pore structures, *Int. J. Heat Mass Transf.* 115 (Dec. 2017) 439–449, <https://doi.org/10.1016/j.jheatmasstransfer.2017.08.066>.
- [19] K. H. Chu, R. Enright, E. N. Wang, Structured surfaces for enhanced pool boiling heat transfer, *Appl Phys Lett*, vol. 100, no. 24, 2012, doi: 10.1063/1.4724190.
- [20] M.M. Rahman, E. Ölçeroglu, M. McCarthy, Role of wickability on the critical heat flux of structured superhydrophilic surfaces, *Langmuir* 30 (37) (2014) 11225–11234, <https://doi.org/10.1021/la5030923>.
- [21] M. Bongarala, H. Hu, J. A. Weibel, S.V. Garimella, Microlayer evaporation governs heat transfer enhancement during pool boiling from microstructured surfaces, *Appl Phys Lett*, vol. 221602, no. May, 2022, doi: 10.1063/5.0090156.
- [22] G. Pi, D. Deng, L. Chen, X. Xu, C. Zhao, Pool boiling performance of 3D-printed reentrant microchannels structures, *Int J Heat Mass Transf* 156 (2020) Aug.
- [23] B. Bon, J.F. Klausner, E. McKenna, The Hoodoo: A new surface structure for enhanced boiling heat transfer, *J. Therm. Sci. Eng. Appl.* 5 (1) (2013) 1–11, <https://doi.org/10.1115/1.4007439>.
- [24] A. Elkholy, R. Kempers, Enhancement of pool boiling heat transfer using 3D-printed polymer fixtures, *Exp. Therm Fluid Sci.* 114 (October) (2020), <https://doi.org/10.1016/j.expthermflusci.2020.110056>.
- [25] Q.N. Pham, S. Zhang, S. Hao, K. Montazeri, C.H. Lin, J. Lee, A. Mohraz, Y. Won, Boiling heat transfer with a well-ordered microporous architecture, *ACS Appl. Mater. Interfaces* 12 (16) (2020) 19174–19183, <https://doi.org/10.1021/acsami.0c01113>.
- [26] S. Mori, K. Okuyama, Enhancement of the critical heat flux in saturated pool boiling using honeycomb porous media, *Int. J. Multiph. Flow* 35 (10) (Oct. 2009) 946–951, <https://doi.org/10.1016/j.ijmultiphaseflow.2009.05.003>.
- [27] D.C. Mo, S. Yang, J.L. Luo, Y.Q. Wang, S.S. Lyu, Enhanced pool boiling performance of a porous honeycomb copper surface with radial diameter gradient, *Int. J. Heat Mass Transf.* 157 (2020), 119867, <https://doi.org/10.1016/j.jheatmasstransfer.2020.119867>.
- [28] Q. Li, Z. Lan, J. Chun, R. Wen, X. Ma, Composite porous surfaces of microcavities for enhancing boiling heat transfer, *Int. J. Heat Mass Transf.* 177 (2021), 121513, <https://doi.org/10.1016/j.jheatmasstransfer.2021.121513>.
- [29] S. Jun, J. Kim, D. Son, H.Y. Kim, S.M. You, Enhancement of pool boiling heat transfer in water using sintered copper microporous coatings, *Nucl. Eng. Technol.* 48 (4) (Aug. 2016) 932–940, <https://doi.org/10.1016/j.net.2016.02.018>.
- [30] J.H. Kim, A. Gurung, M. Amaya, S.M. Kwark, S.M. You, Microporous coatings to maximize pool boiling heat transfer of saturated R-123 and water, *J. Heat Transfer* 137 (8) (2015) 1–7, <https://doi.org/10.1115/1.4030245>.
- [31] S. Sarangi, J.A. Weibel, S.V. Garimella, Effect of particle size on surface-coating enhancement of pool boiling heat transfer, *Int. J. Heat Mass Transf.* 81 (2015) 103–113, <https://doi.org/10.1016/j.jheatmasstransfer.2014.09.052>.
- [32] R. Chen, M.C. Lu, V. Srinivasan, Z. Wang, H.H. Cho, A. Majumdar, Nanowires for enhanced boiling heat transfer, *Nano Lett.* 9 (2) (2009) 548–553, <https://doi.org/10.1021/nl8026857>.
- [33] M.-C. Lu, R. Chen, V. Srinivasan, V.P. Carey, A. Majumdar, Critical heat flux of pool boiling on Si nanowire array-coated surfaces, *Int. J. Heat Mass Transf.* 54 (25–26) (Dec. 2011) 5359–5367, <https://doi.org/10.1016/j.jheatmasstransfer.2011.08.007>.
- [34] D. Il Shim, G. Choi, N. Lee, T. Kim, B. S. Kim, H. H. Cho, Enhancement of pool boiling heat transfer using aligned silicon nanowire arrays, *ACS Appl Mater Interfaces* 9(20) (2017) 17595–17602, doi: 10.1021/acsami.7b01929.
- [35] N.S. Dhillon, J. Buongiorno, K.K. Varanasi, Critical heat flux maxima during boiling crisis on textured surfaces, *Nat. Commun.* 6 (1) (Nov. 2015) 8247, <https://doi.org/10.1038/ncomms9247>.
- [36] C. Li, Z. Wang, P.I. Wang, Y. Peles, N. Koratkar, G.P. Peterson, Nanostructured copper interfaces for enhanced boiling, *Small* 4 (8) (2008) 1084–1088, <https://doi.org/10.1002/smll.200700991>.
- [37] Y. Song, C.D. Díaz-Marín, L. Zhang, H. Cha, Y. Zhao, E.N. Wang, Three-tier hierarchical structures for extreme pool boiling heat transfer performance, *Adv. Mater.* 2200899 (2022) 2200899, <https://doi.org/10.1002/adma.202200899>.
- [38] D. Lee, N. Lee, D. Il Shim, B.S. Kim, H.H. Cho, Enhancing thermal stability and uniformity in boiling heat transfer using micro-nano hybrid surfaces (MNHS), *Appl. Therm. Eng.* 130 (2018) 710–721, <https://doi.org/10.1016/j.applthermaleng.2017.10.144>.
- [39] K.H. Chu, Y.S. Joung, R. Enright, C.R. Buie, E.N. Wang, Hierarchically structured surfaces for boiling critical heat flux enhancement, *Appl. Phys. Lett* 102(15) (2013), doi: 10.1063/1.4801811.
- [40] S. Xie, M. Jiang, H. Kong, Q. Tong, J. Zhao, An experimental investigation on the pool boiling of multi-orientated hierarchical structured surfaces, *Int. J. Heat Mass Transf.* 164 (2021), 120595, <https://doi.org/10.1016/j.jheatmasstransfer.2020.120595>.
- [41] J. Li, W. Fu, B. Zhang, G. Zhu, N. Miljkovic, Ultrascable three-tier hierarchical nanoengineered surfaces for optimized boiling, *ACS Nano* 13 (2019) 14080–14093, <https://doi.org/10.1021/acsnano.9b06501>.
- [42] X. Dai, X. Huang, F. Yang, X. Li, J. Sightler, Y. Yang, C. Li, Enhanced nucleate boiling on horizontal hydrophobic-hydrophilic carbon nanotube coatings, *Appl. Phys. Lett.* 102(16) (2013) doi: 10.1063/1.4802804.
- [43] C. Zhang, L. Zhang, H. Xu, P. Li, B. Qian, Performance of pool boiling with 3D grid structure manufactured by selective laser melting technique, *Int. J. Heat Mass Transf.* 128 (2019) 570–580, <https://doi.org/10.1016/j.jheatmasstransfer.2018.09.021>.
- [44] A.M. Sharifzadeh, H. Moghadasi, H. Saffari, M. Delpisheh, Experimental investigation of pool boiling heat transfer enhancement using electrodeposited open-cell metal foam, *Int. J. Therm. Sci.* 176 (2022), 107536, <https://doi.org/10.1016/j.ijthermalsci.2022.107536>.
- [45] S. Zhang, G. Chen, X. Jiang, Y. Li, S. Waqar, A. Shah, Y. Tang, Z. Wang, C. Pan, Hierarchical gradient mesh surfaces for superior boiling heat transfer, *Appl. Therm. Eng.* (2022), 119513, <https://doi.org/10.1016/j.applthermaleng.2022.119513>.
- [46] G. Chen, N. Chukwunenye, G.F. Jones, C.H. Li, Biomimetic structures by leaf vein growth mechanism for pool boiling heat transfer enhancements, *Int. J. Heat Mass Transf.* 155 (2020), 119699, <https://doi.org/10.1016/j.jheatmasstransfer.2020.119699>.
- [47] Y. Yang, X. Ji, J. Xu, Pool boiling heat transfer on copper foam covers with water as working fluid, *Int. J. Therm. Sci.* 49 (7) (Jul. 2010) 1227–1237, <https://doi.org/10.1016/j.ijthermalsci.2010.01.013>.
- [48] Z.G. Xu, Z.G. Qu, C.Y. Zhao, W.Q. Tao, Pool boiling heat transfer on open-celled metallic foam sintered surface under saturation condition, *Int. J. Heat Mass Transf.* 54 (17–18) (Aug. 2011) 3856–3867, <https://doi.org/10.1016/j.jheatmasstransfer.2011.04.043>.
- [49] J. Xu, X. Ji, W. Zhang, G. Liu, Pool boiling heat transfer of ultra-light copper foam with open cells, *Int. J. Multiph. Flow* 34 (11) (Nov. 2008) 1008–1022, <https://doi.org/10.1016/j.ijmultiphaseflow.2008.05.003>.
- [50] A.M. Sharifzadeh, H. Moghadasi, H. Shakeri, H. Saffari, Influence of bubble departure control on nucleate pool boiling heat transfer of electrodeposited copper foam: experiments and correlation, *Int. Commun. Heat Mass Transfer* 138 (2022), 106381, <https://doi.org/10.1016/j.icheatmasstransfer.2022.106381>.
- [51] Z.G. Qu, Z.G. Xu, C.Y. Zhao, W.Q. Tao, Experimental study of pool boiling heat transfer on horizontal metallic foam surface with crossing and single-directional V-shaped groove in saturated water, *Int. J. Multiph. Flow* 41 (May 2012) 44–55, <https://doi.org/10.1016/j.ijmultiphaseflow.2011.12.007>.
- [52] H. Hu, Y. Zhao, Z. Lai, C. Hu, Influence of surface wettability on pool boiling heat transfer on metal foam covers, *Int. J. Therm. Sci.* 168(2020) 107069, doi: 10.1016/j.ijthermalsci.2021.107069.
- [53] J. Shi, H. Du, Z. Chen, S. Lei, Review of phase change heat transfer enhancement by metal foam, *Appl. Therm. Eng.* 219 (2023), 119427, <https://doi.org/10.1016/j.applthermaleng.2022.119427>.
- [54] M. Calati, G. Righetti, L. Doretto, C. Zilio, G.A. Longo, K. Hooman, S. Mancin, Water pool boiling in metal foams: from experimental results to a generalized model based on artificial neural network, *Int. J. Heat Mass Transf.* 176 (2021), 121451, <https://doi.org/10.1016/j.jheatmasstransfer.2021.121451>.
- [55] C. Huang, H. Wang, E. Lichtouse, Y. Tang, H. Xiang, Experimental study of multilayer gradient copper foam effect on pool boiling heat transfer performance and gas-liquid behavior characteristics, *Int. J. Therm. Sci.* 183(2022) 107856, doi: 10.1016/j.ijthermalsci.2022.107856.
- [56] Z.G. Xu, Z.G. Qu, C.Y. Zhao, W.Q. Tao, Experimental correlation for pool boiling heat transfer on metallic foam surface and bubble cluster growth behavior on grooved array foam surface, *Int. J. Heat Mass Transf.* 77 (2014) 1169–1182, <https://doi.org/10.1016/j.jheatmasstransfer.2014.06.037>.
- [57] A. Bharadwaj, R.D. Misra, A theoretical model to predict pool boiling critical heat flux for micro/nano-structured surfaces, *J. Heat Transfer* 144 (10) (2022) 1–9, <https://doi.org/10.1115/1.4054899>.
- [58] S.G. Kandlikar, Controlling bubble motion over heated surface through evaporation momentum force to enhance pool boiling heat transfer, *Appl. Phys. Lett.* 102 (5) (2013) pp, <https://doi.org/10.1063/1.4791682>.
- [59] Y. Song, L. Zhang, C.D. Díaz-Marín, S.S. Cruz, E.N. Wang, Unified descriptor for enhanced critical heat flux during pool boiling of hemi-wicking surfaces, *Int. J. Heat Mass Transf.* 183 (2022), 122189, <https://doi.org/10.1016/j.jheatmasstransfer.2021.122189>.
- [60] H. Hu, J.A. Weibel, S.V. Garimella, A coupled wicking and evaporation model for prediction of pool boiling critical heat flux on structured surfaces, *Int. J. Heat Mass Transf.* 136 (2019) 373–382, <https://doi.org/10.1016/j.jheatmasstransfer.2019.03.005>.
- [61] M. Winter, J.A. Weibel, The capillary length scale determines the influence of bubble-fin interactions and prediction of pool boiling from heat sinks, *Int. J. Heat Mass Transf.* 202 (2023), 123727, <https://doi.org/10.1016/j.jheatmasstransfer.2022.123727>.
- [62] H.S. Ahn, H.J. Jo, S.H. Kang, M.H. Kim, Effect of liquid spreading due to nano/microstructures on the critical heat flux during pool boiling, *Appl. Phys. Lett.* 98 (7) (2011) 98–101, <https://doi.org/10.1063/1.3555430>.
- [63] S. Poudel, A. Zou, S.C. Maroo, Wicking in cross-connected buried nanochannels, *J. Phys. Chem. C* 123 (38) (2019) 1–6, <https://doi.org/10.1021/acs.jpcc.9b06666>.
- [64] J. Li, Y. Zhao, J. Ma, W. Fu, X. Yan, K.F. Rabbi, N. Miljkovic, Superior antidegradation hierarchical nanoengineered wicking surfaces for boiling enhancement, *Adv. Funct. Mater.* (2021) 2108836, <https://doi.org/10.1002/adfm.202108836>.
- [65] J. Chun, C. Xu, Y. Zhang, Q. Li, R. Wen, X. Ma, Fast capillary wicking on hierarchical copper nanowired surfaces with interconnected V-grooves: implications for thermal management, *ACS Appl Nano Mater* 4 (5) (2021) 5360–5371, <https://doi.org/10.1021/acsnm.1c00691>.
- [66] H.S. Ahn, G. Park, J.M. Kim, J. Kim, M.H. Kim, The effect of water absorption on critical heat flux enhancement during pool boiling, *Exp. Therm Fluid Sci.* 42 (2012) 187–195, <https://doi.org/10.1016/j.expthermflusci.2012.05.005>.
- [67] T.P. Allred, J.A. Weibel, S.V. Garimella, A wettability metric for characterization of capillary flow on textured superhydrophilic surfaces, *Langmuir* 33 (32) (2017) 7847–7853, <https://doi.org/10.1021/acs.langmuir.7b01522>.
- [68] G. Yang, J. Liu, X. Cheng, Y. Wang, X. Chu, S. Mukherjee, A. Terzis, A. Schneemann, W. Li, J. Wu, R.A. Fischer, A superhydrophilic metal-organic

- framework thin film for enhancing capillary-driven boiling heat transfer, *J Mater Chem A Mater* 9 (45) (2021) 25480–25487, <https://doi.org/10.1039/d1ta06826a>.
- [69] R. Xiao, R. Enright, E.N. Wang, Prediction and optimization of liquid propagation in micropillar arrays, *Langmuir* 26 (19) (2010) 15070–15075, <https://doi.org/10.1021/la102645u>.
- [70] A. Rokoni, D.-O. Kim, Y. Sun, Micropattern-controlled wicking enhancement in hierarchical micro/nanostructures, *Soft Matter* 15 (32) (2019) 6518–6529, <https://doi.org/10.1039/c9sm01055f>.
- [71] D.I. Yu, S. Doh, H.J. Kwak, J. Hong, N.P. Sapkal, M.H. Kim, Direct visualization of the behavior and shapes of the nanoscale menisci of an evaporating water droplet on a hydrophilic nanotextured surface via high-resolution synchrotron X-ray imaging, *Langmuir* 35 (19) (2019) 6460–6467, <https://doi.org/10.1021/acs.langmuir.8b04109>.
- [72] D.I. Yu, H.J. Kwak, H. Noh, H.S. Park, K. Fezzaa, M.H. Kim, Synchrotron x-ray imaging visualization study of capillary-induced flow and critical heat flux on surfaces with engineered micropillars, *Sci. Adv.* 4 (2) (2018) 37–42, <https://doi.org/10.1126/sciadv.1701571>.
- [73] J.P. McHale, S.V. Garimella, Nucleate boiling from smooth and rough surfaces - Part 2: Analysis of surface roughness effects on nucleate boiling, *Exp. Therm. Fluid Sci.* 44 (2013) 439–455, <https://doi.org/10.1016/j.expthermflusci.2012.08.005>.
- [74] Y. Li, Y. Li, W. Jiao, X. Chen, G. Lu, Manipulating the heat transfer of pool boiling by tuning the bubble dynamics with mixed wettability surfaces, *Int. J. Heat Mass Transf.* 170 (2021), 120996, <https://doi.org/10.1016/j.ijheatmasstransfer.2021.120996>.
- [75] S.J. Thiagarajan, R. Yang, C. King, S. Narumanchi, Bubble dynamics and nucleate pool boiling heat transfer on microporous copper surfaces, *Int. J. Heat Mass Transf.* 89 (2015) 1297–1315, <https://doi.org/10.1016/j.ijheatmasstransfer.2015.06.013>.
- [76] T.P. Allred, J.A. Weibel, S.V. Garimella, The petal effect of parahydrophobic surfaces offers low receding contact angles that promote effective boiling, *Int. J. Heat Mass Transf.* 135 (2019) 403–412, <https://doi.org/10.1016/j.ijheatmasstransfer.2019.02.002>.
- [77] M. Bucci, J. Buongiorno, M. Bucci, The not-so-subtle flaws of the force balance approach to predict the departure of bubbles in boiling heat transfer, *Phys. Fluids* 33 (1) (2021) pp, <https://doi.org/10.1063/5.0036956>.
- [78] Y. Hsu and R. W. Graham, "AN ANALYTICAL AND EXPERIMENTAL STUDY OF THE THERMAL BOUNDARY LAYER AND EBULLITION CYCLE IN NUCLEATE BOILING," United States, 1961. [Online]. Available: <https://www.osti.gov/biblio/4017144>.
- [79] Y.H. Chang, Y.M. Ferng, Experimental investigation on bubble dynamics and boiling heat transfer for saturated pool boiling and comparison data with previous works, *Appl. Therm. Eng.* 154 (February) (2019) 284–293, <https://doi.org/10.1016/j.applthermaleng.2019.03.092>.
- [80] L. Dong, X. Quan, P. Cheng, An experimental investigation of enhanced pool boiling heat transfer from surfaces with micro/nano-structures, *Int. J. Heat Mass Transf.* 71 (2014) 189–196, <https://doi.org/10.1016/j.ijheatmasstransfer.2013.11.068>.
- [81] H. Chi-Yeh, P. Griffith, The mechanism of heat transfer in nucleate pool boiling-Part I. Bubble initiation, growth and departure, *Int. J. Heat Mass Transf.* 8 (6) (1965) 887–904, [https://doi.org/10.1016/0017-9310\(65\)90073-6](https://doi.org/10.1016/0017-9310(65)90073-6).
- [82] Y. Park, H. Kim, J. Kim, H. Kim, Measurement of liquid-vapor phase distribution on nano- and microstructured boiling surfaces, *Int. J. Multiph. Flow* 81 (2016) 67–76, <https://doi.org/10.1016/j.ijmultiphaseflow.2016.01.007>.
- [83] H. Kim, Y. Park, J. Buongiorno, Measurement of wetted area fraction in subcooled pool boiling of water using infrared thermography, *Nucl. Eng. Des.* 264 (2013) 103–110, <https://doi.org/10.1016/j.nucengdes.2013.07.002>.
- [84] D.E. Kim, J. Park, Experimental study of critical heat flux in pool boiling using visible-ray optics, *Int. J. Heat Mass Transf.* 169 (2021), 120937, <https://doi.org/10.1016/j.ijheatmasstransfer.2021.120937>.
- [85] D.E. Kim, J. Song, H. Kim, Simultaneous observation of dynamics and thermal evolution of irreversible dry spot at critical heat flux in pool boiling, *Int. J. Heat Mass Transf.* 99 (2016) 409–424, <https://doi.org/10.1016/j.ijheatmasstransfer.2016.04.009>.
- [86] J.P. McHale, S.V. Garimella, Bubble nucleation characteristics in pool boiling of a wetting liquid on smooth and rough surfaces, *Int. J. Multiph. Flow* 36 (4) (2010) 249–260, <https://doi.org/10.1016/j.ijmultiphaseflow.2009.12.004>.
- [87] Y. Suh, J. Lee, P. Simadiri, X. Yan, S. Sett, L. Li, K.F. Rabbi, N. Miljkovic, Y. Won, A Deep Learning Perspective on Dropwise Condensation, *Adv. Sci.* 2101794 (2021) 2101794, <https://doi.org/10.1002/advs.202101794>.
- [88] S. Ridwan, M. McCarthy, Nanostructure-supported evaporation underneath a growing bubble, *ACS Appl. Mater. Interfaces* 11 (13) (2019) 12441–12451, <https://doi.org/10.1021/acsami.8b21260>.
- [89] M. Bongarala, H. Hu, J.A. Weibel, S.V. Garimella, A figure of merit to characterize the efficacy of evaporation from porous microstructured surfaces, *Int. J. Heat Mass Transf.* 182 (2021), <https://doi.org/10.1016/j.ijheatmasstransfer.2021.121964>.
- [90] J. Li, G. Zhu, D. Kang, W. Fu, Y. Zhao, N. Miljkovic, Endoscopic visualization of contact line dynamics during pool boiling on capillary-activated copper microchannels, *Adv. Funct. Mater.* 31 (4) (2021) 1–9, <https://doi.org/10.1002/adfm.202006249>.
- [91] J. Li, D. Kang, K.F. Rabbi, W. Fu, X. Yan, X. Fang, L. Fan, N. Miljkovic, Liquid film-induced critical heat flux enhancement on structured surfaces, *Sci. Adv.* 7 (26) (2021) 1–13, <https://doi.org/10.1126/sciadv.abg4537>.
- [92] H. Hu, J.A. Weibel, S.V. Garimella, Role of nanoscale roughness in the heat transfer characteristics of thin film evaporation, *Int. J. Heat Mass Transf.* 150 (2020), 119306, <https://doi.org/10.1016/j.ijheatmasstransfer.2020.119306>.
- [93] K. Nishant Ranjan Sinha, D. Ranjan, M. Q. Raza, N. Kumar, S. Kaner, A. Thakur, R. Raj, In-situ acoustic detection of critical heat flux for controlling thermal runaway in boiling systems, *Int. J. Heat Mass Transf.* 138 (2019) 135–143, doi: 10.1016/j.ijheatmasstransfer.2019.04.029.
- [94] H. Pandey, W. Waldo, H. Hu, Non-intrusive cooling system fault detection and diagnostics using acoustic emission, in: *Proceedings of the ASME 2022 Heat Transfer Summer Conference*, 2022, pp. HT2022–85429, <https://doi.org/10.1115/HT2023-106015>.
- [95] Y. Ueki, K. Ara, Proof of concept of acoustic detection of boiling inception and state transition using deep neural network, *Int. Commun. Heat Mass Transfer* 129 (2021), 105675, <https://doi.org/10.1016/j.icheatmasstransfer.2021.105675>.
- [96] S. Bin Seo, I.C. Bang, Acoustic analysis on the dynamic motion of vapor-liquid interface for the identification of boiling regime and critical heat flux, *Int J Heat Mass Transf* 131 (2019) 1138–1146, <https://doi.org/10.1016/j.ijheatmasstransfer.2018.11.136>.
- [97] A. Negi, A.M. Rishi, S.G. Kandlikar, Effect of heat flux on bubble coalescence phenomena and sound signatures during pool boiling, *J. Heat Transfer* 143 (5) (2021) pp, <https://doi.org/10.1115/1.4050088>.
- [98] S. Barathula, J.K.K. Alapati, K. Srinivasan, Investigation of acoustic spectral variations in the pool boiling regimes of water on wire heater, *Appl. Therm. Eng.* (2023), 120281, <https://doi.org/10.1016/j.applthermaleng.2023.120281>.
- [99] D.Y. Lim, J.Y. Kim, D.H. Lee, K.M. Kim, I.C. Bang, Identifying heat transfer regimes by acoustic analysis in pool and flow boiling, 2020, pp. 9–11.
- [100] C. Dunlap, H. Pandey, E. Weems, H. Hu, Nonintrusive heat flux quantification using acoustic emissions during pool boiling, *Appl Therm Eng.* vol. 228, no. December 2022, p. 120558, 2023, doi: 10.1016/j.applthermaleng.2023.120558.
- [101] A.H. Mehrabi, S.K. Conlin, T.I. Hollis, B.S. Gattis, N. Weker, R.H. Coridan, Electrochemical control of the morphology and functional properties of hierarchically structured, dendritic Cu surfaces, *Energ. Technol.* (2022), <https://doi.org/10.1002/ente.202201124>.
- [102] E. Achilli, A. Vertova, A. Visibile, C. Locatelli, A. Minguzzi, S. Rondinini, P. Ghigna, Structure and stability of a Copper(II) Lactate complex in alkaline solution: a case study by energy-dispersive X-ray absorption spectroscopy, *Inorg. Chem.* 56 (12) (2017) 6982–6989, <https://doi.org/10.1021/acs.inorgchem.7b00553>.
- [103] S.H. Baek, K. Wu, H.S. Shim, D.H. Lee, J.G. Kim, D.H. Hur, Acoustic emission monitoring of water boiling on fuel cladding surface at 1 bar and 130 bar, *Measurement (Iond)* 109 (2017) 18–26, <https://doi.org/10.1016/j.measurement.2017.05.042>.
- [104] Q. Zhang, K. Dong, Z. Huang, J. Wang, Y. Yang, Mode identification of pool boiling of water by acoustic emission, *CIESC Journal* 64 (10) (2013) pp.
- [105] J. Marsh, C. Dunlap, S. Pierson, and H. Hu, "Introducing LabVIEW and Arduino as Data Acquisition System Alternatives," in: 2022 ASME Midwest Section Conference, Paper 3279, 2022.
- [106] N. Fries, M. Dreyer, The transition from inertial to viscous flow in capillary rise, *J. Colloid Interface Sci.* 327 (1) (2008) 125–128, <https://doi.org/10.1016/j.jcis.2008.08.018>.
- [107] M. Borumand, T. Lee, G. Hwang, Enhanced wickability of single-columnar, non-uniform pore-size wick using Lattice Boltzmann Method, *Comput. Fluids* 238 (2022), 105376, <https://doi.org/10.1016/j.compfluid.2022.105376>.
- [108] R.I. Vachon, G.H. Nix, G.E. Tanger, Evaluation of constants for the Rohsenow pool-boiling correlation, *J. Heat Transfer* 90 (2) (1968) 239–246.
- [109] S. Fan, L. Jiao, K. Wang, F. Duan, Pool boiling heat transfer of saturated water on rough surfaces with the effect of roughening techniques, *Int. J. Heat Mass Transf.* 159 (2020), 120054, <https://doi.org/10.1016/j.ijheatmasstransfer.2020.120054>.
- [110] H. Pandey, C. Dunlap, A. Williams, J. Marsh, H. Hu, Multimodal characterization of steady-state and transient boiling heat transfer, in: *Proceedings of the ASME 2023 Heat Transfer Summer Conference*, 2023, pp. HT2023-106015.
- [111] K. Nishant Ranjan Sinha, D. Ranjan, N. Kumar, M. Qaisar Raza, R. Raj, Simultaneous audio-visual-thermal characterization of transition boiling regime, *Exp. Therm. Fluid Sci.* 118 (2020) 110162, doi: 10.1016/j.expthermflusci.2020.110162.
- [112] K. Nishant Ranjan Sinha, D. Ranjan, N. Kumar, M. Qaisar Raza, R. Raj, Simultaneous audio-visual-thermal characterization of transition boiling regime, *Exp. Therm. Fluid Sci.* 118 (2020) 110162, doi: 10.1016/j.expthermflusci.2020.110162.
- [113] J.D. Bernardin, I. Mudawar, The Leidenfrost point: Experimental study and assessment of existing models, *J. Heat Transfer* 121 (4) (1999) 894–903, <https://doi.org/10.1115/1.2826080>.
- [114] J. Schindelin, I. Arganda-Carreras, E. Frise, V. Kaynig, M. Longair, T. Pietzsch, S. Preibisch, C. Rueden, S. Saalfeld, B. Schmid, J.Y. Tinevez, D.J. White, V. Hartenstein, K. Eliceiri, P. Tomancak, A. Cardona, Fiji: An open-source platform for biological-image analysis, *Nat. Methods* 9 (7) (2012) 676–682, <https://doi.org/10.1038/nmeth.2019>.
- [115] M. Matković, B. Končar, Bubble departure diameter prediction uncertainty, *Sci. Technol. Nucl. Instal.* 2012 (2012), <https://doi.org/10.1155/2012/863190>.
- [116] S. Barathula, K. Srinivasan, Review on research progress in boiling acoustics, *Int. Commun. Heat Mass Transfer* 139 (2022), 106465, <https://doi.org/10.1016/j.icheatmasstransfer.2022.106465>.

- [117] S. Barathula, J.K.K. Alapati, K. Srinivasan, Investigation of acoustic spectral variations in the pool boiling regimes of water on wire heater, *Appl. Therm. Eng.* 226 (2023), 120281, <https://doi.org/10.1016/j.applthermaleng.2023.120281>.
- [118] S. Husin, An experimental investigation into the correlation between acoustic emission (AE) and bubble dynamics, Ph.D. Thesis, Cranfield University, 2011.
- [119] C. Mira-Hernández, J.A. Weibel, P.P. Vlachos, S.V. Garimella, Three-dimensional liquid-vapor interface reconstruction from high-speed stereo images during pool boiling, *Int. J. Heat Mass Transf.* 136 (2019) 265–275, <https://doi.org/10.1016/j.ijheatmasstransfer.2019.02.077>.